



Elliott Sound Products

The Design of Heatsinks

## The Design of Heatsinks

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### Introduction

This article is essential reading for anyone who uses heatsinks in audio, and the same principles apply for power control, RF and CPU cooling, as well as many other applications in electronics. There are many misunderstandings and misconceptions about transistor mounting. What are the essential requirements? Which things make the biggest difference? These questions may not have occurred to many readers, but they are very important to ensure the long term reliability of your projects. A point I have made in several articles and projects needs to be reiterated here ...

### *There Is No Such Thing As A Heatsink That's Too Big*

The bigger and better the heatsink you use, the lower the operating temperature of the devices

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mounted on it, provided you take great care to ensure that the interface between the device(s) and heatsink are as good as you can achieve. Much of this article concentrates on the mounting techniques, and if you get that wrong, it won't matter how good your heatsink is. If the devices are hot and the heatsink is cold, then you haven't mounted the devices properly, or you used an inappropriate mounting technique (such as silicone washers for a transistor dissipating high power).

Contrary to what is believed by some, the job of the heatsink is *not* to get hot. An ideal heatsink would remain at ambient temperature (or less), providing the lowest possible operating temperature for the devices being cooled. This assumes that everything else is right, as described in this article. From an *accounting* perspective, the heatsink should be as small as possible to minimise cost, but the accountant(s) won't have to repair gear that failed because it got too hot.



**Please** ... read all of this article - not just the bit you think you need. You would be surprised how much I learned while researching and writing this material, and I have had a lot of additional information supplied by readers (you know who you are, and thank you for your input). The material here is the result of a lot of hard work, research, personal experience and (what should be) common knowledge.

The design of heatsinks, or to be more exact for most builders, the selection of a suitable heatsink, is not difficult once the basics have been mastered. Terms such as 'thermal resistance' and '°C / Watt' are a little daunting for the uninitiated, and the purpose of this article is to explain how thermal transfer works, from the transistor die until it is finally dispersed into the atmosphere.

This sort of information seems to be unusually difficult to obtain, either on the web or elsewhere, and I have had great difficulty in getting thermal resistance data from anywhere - although as you shall see as you read on, I have managed it. Not that this article is the 'last word' on heatsinks - far from it, but it is a collection of some of the most useful information I have been able to gather thus far.

Note that in this article, the 'transistor' can be a bipolar transistor, MOSFET, TRIAC, CPU, light emitting diode (including laser diodes) or any other semiconductor or passive device (including rectifier diodes, ICs of all kinds, high power resistors, etc.) that is mounted in a plastic, ceramic and/or metal case which in turn must be mounted on a heatsink of some description.

A whole new field is opening up now, and that's providing cooling for LEDs of ever increasing power. Because these form parts of lighting systems that are visible to all, heatsinks for LED lighting applications commonly break all the rules you'll read about in this article. This is so they will look 'nice' so people will buy them. Unfortunately, by breaking the rules, the heatsinks and LEDs often run far hotter than we might prefer, which shortens the life of the LEDs and the associated power supply that drives them. Some designs have even included tiny little fans inside the lamp - the reliability of a 20mm diameter fan exposed to ceiling-space dust is rather suspect at best.

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***The reliability and longevity of any semiconductor device is inversely proportional to the junction temperature. The hotter the junction, the shorter the expected life. Reducing the junction temperature by 10°C will result in approximately double the expected life of the component. The converse is also true! A worthwhile increase in reliability and component life can be achieved by a relatively small reduction in operating temperature.***

***However, life is actually somewhat more complex than this simple 'rule' implies. If you're lucky, the manufacturer will provide details on temperature vs. lifespan, but it's not something that you can rely on (it's very rare to see this information). As a general rule it's worth considering though, even if it's not particularly accurate.***

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The whole process of removing heat from a transistor's active area (the die or 'junction') involves many separate thermal transfers, and we shall examine each in turn. In order to see a meaningful result, we need a heat input value and an ambient air temperature. From these two basic elements, the entire heat flow can be established.

Towards this end, an electrical analogue may be drawn, showing the thermal generator as a current source, thermal resistance as a resistor and the thermal inertia (or transient capacity) of the various materials as a capacitor. The transient capacity is the ability of any material to absorb a quantity of heat for a short time, before its temperature rises noticeably. This happens in the same way as the voltage rises in a capacitor both with 'transient' events (small effect) and when the supply current is maintained.



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Thermal inertia is especially important when the dissipated power is not constant (this is almost always the case with audio amplifiers).

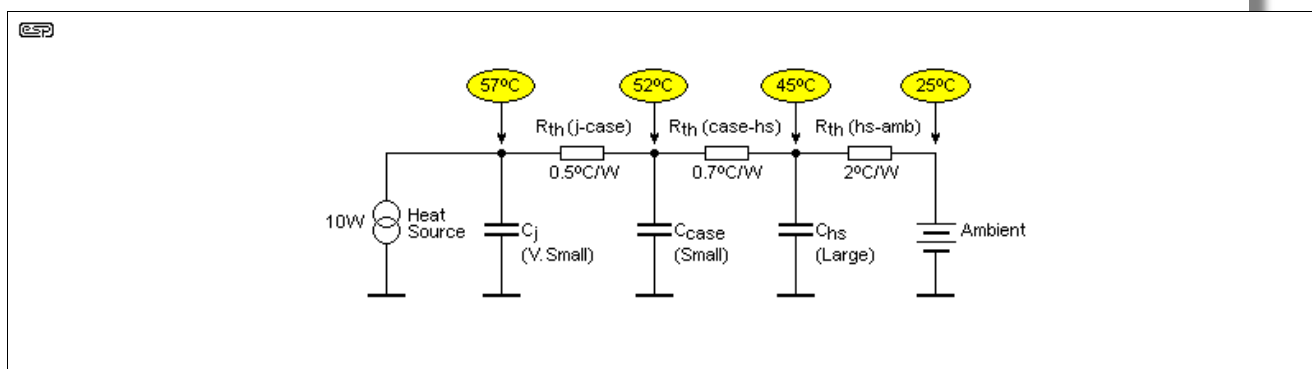


Figure 1 - Heat Flow From Generator To Ambient

From Figure 1, we can see the thermal resistance ( $R_{th}$ ) from junction to case, the thermal inertia of the junction (this is very small), the thermal inertia of the case itself (again, not large), the thermal resistance from the case to the heatsink, thermal inertia of the heatsink (this might be very large) and finally the thermal resistance from the heatsink to ambient. Ambient temperature is shown as a 'voltage' source, and the heatsink *cannot* be at a lower temperature than that without using a refrigeration system (a Peltier device for example).

**'Ambient' means the temperature surrounding the heatsink, *not* room temperature!**

As an example, the heat source shown is 10W, and the thermal gradient across each thermal resistance means that the junction will always be the hottest (57°C), the case will be slightly cooler (52°C), and the heatsink cooler again (45°C). Finally, we reach the ambient temperature (25°C). The total thermal resistance from junction to ambient is 3.2°C/W (the sum of all thermal resistances). It follows that if the ambient is 25°C and the power is 10W, the junction must be at a temperature of ...

$$T_j = \text{Power} \times R_{th} + T_{amb}$$

$$T_j = 10 \times 3.2 + 25 = 57^\circ\text{C}$$

Note (again) that the word 'ambient' does *not* mean the temperature in the room, outside, or anywhere other than immediately adjacent to the heatsink. Any other heat source (including other heatsinks) raises the temperature of its surroundings. This is an easy mistake to make, so it's essential to ensure that you know how the 'typical' temperature near the heatsink itself. This is also influenced by other heatsinks located nearby. For example, think of the ramifications of two high-power amplifiers mounted in a rack, one above the other. Unless a fan is used, the upper heatsink must have a higher 'ambient' temperature than the lower heatsink.

Such measurements can only be made after the equipment has been running for some time, *especially* if the heatsink is not directly exposed to the air in the room. Stacking equipment can mean that the 'ambient' temperature at the top of the stack is much higher than that at the bottom, particularly if the equipment at the bottom of the stack generates significant heat itself. These considerations can easily lead to the temperature next to your heatsink being significantly higher than you may have anticipated.

It's not uncommon to see these thermal resistances written as  $\theta_{jc}$  (thermal resistance, junction to case),  $\theta_{cs}$  (case to (heat)sink), etc. Mostly, you'll be able to figure out what is intended from the context, regardless of the terminology used.

The small thermal inertia values should be ignored, as they will heat up very rapidly, and the heatsink itself will only absorb so much heat before its temperature starts to rise noticeably. A state of equilibrium is required where the heat input equals the heat output, but without the junction temperature reaching a dangerous level - even momentarily. Aiming for the lowest possible junction temperature means that the equipment should have minimal thermal stress.

## 1 - Definitions



There are many definitions used, some are easy to understand without explanation, others less so. One of the most important is the definition of 'ambient temperature' which is the temperature

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immediately adjacent to a heatsink or other hot component. It is *not* the temperature in the room unless the heatsink(s) are exposed to the air in the room with no impediments to free airflow.

- Heat Load (W) - Amount of heat energy produced in watts. Usually defined as average voltage drop across device  $\times$  current flowing through device
- Ambient Temperature ( $T_{amb}$ ) - Temperature of air immediately around Device to be cooled (NOT temperature in the room !)
- Maximum Junction Temperature, ( $T_{jmax}$ ) - The maximum allowable temperature that the device will reach at its junction
- Thermal Resistance ( $R\theta$ ) ( $^{\circ}\text{C}/\text{W}$ ) - Also written as  $R_{th}$  and/ or  $\text{K}/\text{W}$  (Kelvin/watt). The resistance that the Heat Energy meets as it flows from hot (device junction) to cold (ambient air)
- Junction-to-Case Thermal Resistance, ( $R\theta_{jc}$ ), of the electronic device from the silicon junction to the case of the package (supplied by manufacturer)
- $R\theta_{cs}$  - Thermal Resistance of the Thermal Interface Material, (TIM), used between the device and heatsink
- $R\theta_{sa}$  - Thermal Resistance of the heatsink to the ambient (surrounding air).

Many of these are used below, others will be found in publications, data sheets or documents you may find elsewhere. Ultimately, it doesn't matter whether these exact terms are used or not, provided the meaning is clear. You will note that temperature is almost invariably stated in Celsius. Fahrenheit is uncommon and rightly so, because it's at odds with the way that semiconductor devices are specified. It's also a singularly useless way to specify temperature because it doesn't make sense.

There are three ways that heat is 'lost'. Conduction carries heat from a semiconductor die to the case, then to the heatsink via thermal interface material(s). Conduction also spreads the heat from the area of contact across the heatsink body (including fins). Convection (usually air) removes heat from the surface of the heatsink into the surrounding atmosphere. Finally, radiation sends heat as infra-red energy from a hot surface into (relatively) colder external space. Radiation does not rely on air - it even works in a vacuum.

Conduction and convection are the most important, with radiation being the least effective way to remove heat from any mass. It does help though, and this is why most heatsinks are black, as that's the most effective colour for radiation. It's also effective for absorption, so keep heatsinks away from direct sunlight as that will make them get hot (even when equipment is turned off !).

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### 1.1 - Commercial Simulation

Because heatsink design is so complex, and especially where space or cost preclude using the biggest heatsink you can fit, there are several thermal simulation packages available. These are most certainly not for the faint-hearted, and they are both expensive to buy and very difficult to master. Everything that influences heat flow is considered, including air density, boundaries (whether as part of the design or external) and thermal gradients. I recently watched a 'pod cast' which featured a simple design, and the commentator said that on his computer (undoubtedly somewhat better than something that mere mortals could afford), the simple simulation demonstrated takes several *hours* to complete.

With the fairly secure knowledge that few hobbyists will have access to such powerful simulation packages, we are left with the rather tedious process of working out everything manually. We are assisted by looking at commercial products (preferably those that have a good reputation for reliability), to see how much (or how little!) heatsinking is used. This is not an exact science, but with experience it doesn't take very long before we are able to determine what will work, based on fundamental principles.

There are potential traps that are difficult to see when using simple calculations. Of these, using material that is too thin as the heat-spreader is one of the easiest to overlook. The heat spreader can be the base of a commercially produced heatsink, a bracket to which the power devices are attached, or just a simple flat plate heatsink. The latter is more common than you may think, such as regulator ICs or bridge rectifiers mounted directly to the chassis. It's especially important to realise that if the case acting as a heatsink is steel (rather than aluminium for example), heat transfer is woeful, because steel is a very poor conductor. This can be alleviated by using a robust (not less than 2.5mm thick) piece of aluminium to disperse the heat over a wider area. If done properly, this will work well, but it's still sub-optimal.

A commercial simulation package can determine the efficacy of such an approach, but most hobbyists



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(and many manufacturers) will use empirical (i.e. 'trial-and-error') techniques to decide if the proposed solution will work or not. In every case, the idea is to minimise the temperature of semiconductors, because operation at high temperatures causes more rapid degradation of the silicon die, and can cause solder joint fractures due to continual thermal expansion and contraction. This can even happen at the IC level, with BGA ICs being particularly vulnerable. These are very uncommon in home construction because specialised equipment is needed to solder them in place.

Ultimately, there isn't really anything that can be taken for granted. Heat is the enemy of all components (with the possible exception of heating elements 😊). Keeping thermal stress at bay isn't an easy task, and the introduction of surface mounted devices (SMD) and lead-free solder makes it all that much harder to get it right.

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## 2 - Types of Heatsinks

Heat sinks can be classified in terms of manufacturing methods and their final form shapes. The most common types of air-cooled heat sinks include:

- **Stampings** Copper or aluminium sheet metals are stamped into desired shapes. They are used in traditional air cooling of electronic components and offer a low cost solution to low density thermal problems. They are suitable for high volume production, because advanced tooling with high speed stamping lowers costs. Additional labour-saving options such as clips and dry interface materials can be factory applied to help to reduce the board assembly costs.
- **Extrusions** These are by far the most common for power amplifiers, and allow the formation of elaborate (in two dimensions) shapes capable of dissipating large heat loads. They may be cut, machined, and options added. Cross-cutting will produce omni-directional, rectangular pin fin heat sinks, and incorporating serrated fins may improve the performance by approximately 10 to 20%. Extrusion limits, such as the fin height-to-gap ratio and fin thickness, usually dictate the flexibility in design options. Typical fin height-to-gap aspect ratio of up to 6:1 and a minimum fin thickness of about 1.3mm are attainable with standard extrusion processes.
- **Bonded / Fabricated Fins** Most air cooled heat sinks are convection limited, and the overall thermal performance of an air cooled heat sink can often be improved significantly if more surface area can be exposed to the air stream. Such high performance heat sinks utilise thermally conductive aluminium-filled epoxy to bond planar fins onto a grooved extrusion base plate. This process allows for a much greater fin height-to-gap aspect ratio of 20 to 40, with the intent of greatly increasing the cooling capacity without increasing volume requirements. Very tall fins don't always work.
- **Castings** Sand, lost core and die casting processes are available with or without vacuum assistance, in aluminium or (less frequently for audio applications) copper/bronze. This technology is used in high density pin fin heat sinks which provide maximum performance when using forced air cooling. Cast aluminium heatsinks look very good, but are usually less effective than extrusions (due to the alloy used for casting).
- **Folded Fins** Corrugated sheet metal - either aluminium or copper, increases surface area and, hence the volumetric performance. The heat sink is then attached to either a base plate or directly to the heating surface via epoxy adhesive or brazing. It is not suitable for high profile heat sinks, but it allows high performance heat sinks to be fabricated for specific applications.
- **Forced Air** Some heatsinks are designed *specifically* for forced air operation, and they will have very poor performance if used without a fan. Most are extruded aluminium, a few may be copper, and they are fabricated with very close fin spacings, and in some cases will have a series of small channels of various sizes and cross-sections. These are almost always rather expensive, and they are supplied with a high-performance fan. These are available from some of the major suppliers. I can only assume that some people buy them (some are *well* over AU\$500 each, depending on thermal resistance).
- **Heat Pipes** These are now fairly common with high-performance CPUs (central processing units) in computers, especially for gaming or server applications. The heat pipe is actually two heatsinks, joined by a pipe containing a refrigerant. Natural convection is used to move the refrigerant from the hot area (the component to be cooled) and the main heatsink, which may be



mounted some distance away. This arrangement allows a higher performance radiator (the fan-blown part) than could normally be fitted into the confines of a computer case.

- **Flat Sheet** Sometimes, a simple flat sheet is adequate, especially for relatively low heat loads. If thick enough (such as an amplifier case), a flat sheet will be enough for low powered amplifiers (IC power amps for example) used for domestic listening at a 'sensible' output power and normal domestic speakers. It's essential to test the system thoroughly before committing to this approach, as it's not a particularly effective method. Performance is improved if air can circulate through the case, as that allows both sides of the sheet to dissipate heat. The aluminium sheet should normally be at least 2mm thick. Flat sheets are also common for voltage regulators and power amplifier driver stages, where the heat load is no more than 2-3 watts. In such cases, 1mm thick aluminium will usually suffice.

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### 3 - Sample Calculation

Let us assume that the heat generated in the transistor die is 50W, such as might be the case with a Class-A power amplifier operating from a +/-25V supply, and drawing 2A quiescent current. Only half the total supply voltage of 50V is across the transistor in the quiescent or no signal state when the output is at zero volts, so each of the transistors in the output stage will have a voltage of 25V at 2A, or 50W dissipation. That's a total of 100W of heat to dissipate.

An ambient temperature of 25°C is a good starting point, and provides a safety margin for most domestic systems, although as we shall see later on, it is important to design for worst case if reliability is to be maintained. If we were to include a safety margin and allow for ambient temperatures of up to 30°C, in order to dissipate 100W of heat and maintain a transistor case temperature of (say) 60°C (30°C temperature rise), the heatsink and thermal interface combined need to have a total thermal resistance of 0.3°C/W. With 100W dissipated by the transistors, the (theoretical) thermal resistance is simply ...

$R_{th} = \text{temperature rise} / \text{power} \dots$  In this case

$R_{th} = 30^\circ\text{C} / 100\text{W} = 0.3^\circ\text{C/W}$

I put 'theoretical' in quotes because it's far more complex than that. It's necessary to determine all of the parameters that affect heat transfer to get a final figure. The idea of this article is to examine each of these factors so that a final figure can be determined. Even after we're done, there can be other variables that weren't considered, and in many cases an estimate is necessary to ensure that the final design is financially and physically viable.

The remainder of this article discusses all of the factors that influence the outcome. There's nothing hard about it, but it will take the reader into perhaps unfamiliar territory. The processes involved aren't trivial, but nor are they especially difficult.

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The primary aim of the heatsink designer is to ensure that the total thermal resistance is kept to the minimum possible value, and the entire design process looks at thermal resistance as the primary item to be calculated. Only after this has been determined can the actual temperature of the transistor junction be predicted.

To some extent, this article was prompted by a reader of my pages, who complained that transistors in an amp he built ran very hot (mounted on a bracket), while others seemed to run fairly cool. So I got to thinking, didn't I?

How to actually mount transistors did not seem to have achieved much coverage on the web when this article was written (ot a bit better now), and I have seen some absolute drivell spouted by PC types talking about heatsinking Pentium processors - the same principles apply, but the amount of heat and thermal dynamics are very different.

Throughout this article I will often refer to 'aluminium', which in this context means aluminium alloy. Pure aluminium is rarely used, since it is too soft and easily distorted. It is also difficult to machine, drill and tap for mounting screws, because it tends to clog the flutes on the drill bit, and will even snap the drill if great care is not taken.



When drilling or tapping threads into aluminium (or any of its alloys), the use of methylated spirit (denatured alcohol), ethyl alcohol or isopropyl alcohol works very well. So too does WD40 or other

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similar 'water displacement' spray. Alcohol acts as a lubricant and is highly recommended, and it's less messy than any oil-based lubricant. This trick is not well known, but is very effective.



Be warned that methylated spirit ('metho' as it is commonly known in Australia) and other forms of alcohol are all highly flammable, and they burn with an almost invisible - but very hot - flame. Extreme care is required, and use only enough to keep the drill bit wet - the last thing we need is one of my readers burning down the house while drilling a heatsink. Needless to say, I can accept no responsibility for any accident caused by the use of this technique.

On the positive side, metho and other alcohols evaporate quickly and leave no residue. This also means that spontaneous combustion is far less likely than with any oily lubricant. You should *never* just drop oily rags in the dustbin, as it is quite possible for them to burst into flame for no apparent reason.

#### 4 - Measuring Heatsink Thermal Resistance

The most accurate way to determine the thermal resistance of an unknown heatsink is to measure it. The exercise is not trivial though, since you will require a large metal clad resistor having a good flat bottom surface (or you can use transistors), a contact thermometer (a conventional alcohol or mercury in glass thermometer cannot be used), and a suitable low voltage, high current power supply. If you have a large number of heatsinks to test it may be worthwhile to build a dedicated test unit, however this is unlikely for most home constructors.

Software exists to allow simulation of heatsink performance, and this can make it easier to work out what a heatsink is capable of doing. However, high performance thermal modelling software is very expensive and hard to use, and isn't appropriate for casual hobbyist use. If you are planning to use a forced-air solution (using a fan), consider that even the very best modelling software may have an uphill battle dealing with turbulent airflow, and empirical testing is the only way to determine the heatsink's performance.

It is important that the heatsink under test is set up as closely as possible to the way it will be used. There is no point testing a sink just lying on the workbench (for example), as the results will be way off. If a heavy chassis is planned, then attach the heatsink to the chassis or a reasonable facsimile thereof. Ensure that the heating system is in the best possible thermal contact with the heatsink. Thermal compound is essential, and do not use any insulators.

The test is based on knowing the voltage and current you apply to the heatsink heating system (resistors or transistors), and being able to accurately measure the ambient and heatsink temperatures. First, apply a relatively low power to the heater system of your choice, and wait for the heatsink temperature to stabilise - this could take an hour or more. If the heatsink is too hot or too cold the results will be inaccurate, so slowly (in steps) increase power until the heatsink is at approximately the maximum temperature you feel is reasonable (typically around 50-60°C).

Measure the ambient temperature and the heatsink temperature, preferably using the same thermometer. A contact thermometer is essential for the heatsink (again, use thermal compound). Determine the temperature difference (temperature rise) between ambient and heatsink.

Next, determine the power applied to your heating system. Thermal resistance may now be established with some very simple maths ...

You will use the following terms -

- $T_r$  - Temperature rise
- $T_a$  - Ambient temperature (example 22°C)
- $T_h$  - Heatsink temperature (example 54°C)
- $V_h$  - Voltage to heater (example 12V)
- $I_h$  - Current through heater (example 3.5A)
- $P_h$  - Power applied to heatsink
- $R_{th}$  - Thermal resistance (in °C/W) so ...

$$T_r = T_h - T_a = 54 - 22 = 32^\circ\text{C}$$

$$P_h = V_h \times I_h = 12 \times 3.5 = 42\text{W}$$



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$$R_{th} = T_r / P_h = 32 / 42 = 0.76^{\circ}\text{C/W}$$

This is as accurate as you need, and as good as you'll get in real life. To get accurate results is time-consuming, and is not necessary because real-world conditions are often highly unpredictable. Once you do the tests a couple of times you will be able to 'guesstimate' the approximate power handling capacity of a heatsink just by looking at it and checking the manufacturer's data. Bear in mind that few heatsink manufacturers supply the all important temperature rise information, and their figures can be off by 25% in either direction. Depending on how the heatsink is mounted you may get significantly different performance.

While you have the test set up, if you have a small fan handy it's worthwhile to prove a point. Using the same heatsink and heater power, set the fan so it blows air against the heatsink under test. The distance doesn't matter much, and a gap of 10-20mm is of little consequence. Measure the thermal resistance again - it will be significantly lower than the free-air case. Next, reverse the fan so it *sucks* air across the fins. The fan has to be mounted very close to the heatsink for this to work at all. Measure thermal resistance again - you'll find that sucking air across the heatsink is nowhere near as good as blowing air directly onto the fins. See [Fan Cooling](#) below.

## 5 - Transistor Case Styles

There is a vast number of different case styles available. I shall only deal with the most common (as shown below), but the information is equally valid for other case styles. In many cases, an educated guess will be needed to determine case to heatsink thermal resistance if the case used by your device is wildly different from those quoted. Generally, this is based on surface area and the evenness (or otherwise) of pressure distribution directly beneath the internal silicon die. This is a wildly variable quantity, and has been known to cause many a constructor grief, not realising the importance of evenly distributed pressure.

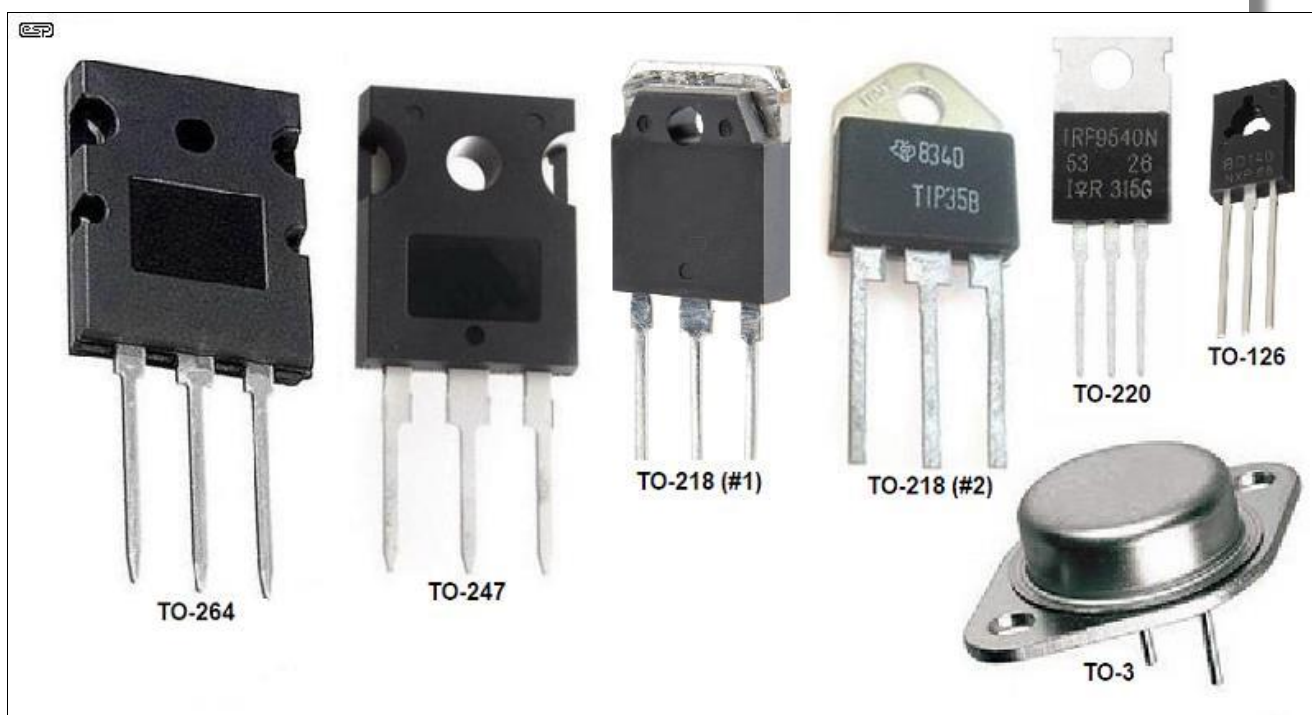


Figure 2 - Common Case Styles

This is only a sample of those currently available (including ICs of various types), but is representative of the most common for transistors, MOSFETs and the like. They are not exactly to scale, but are fairly close (I hope). In most cases, the die can be assumed to be roughly in the middle of the plastic moulding for all plastic encapsulated devices. In the TO-3 package, it is located in the geometric centre of the package (more or less). Other packages also exist (stud-mounted devices for example), and it's not always easy to work out the case-heatsink thermal resistance. The latter is highly dependent on the TIM (thermal interface material), mounting technique and the surface condition of both semiconductor and heatsink.

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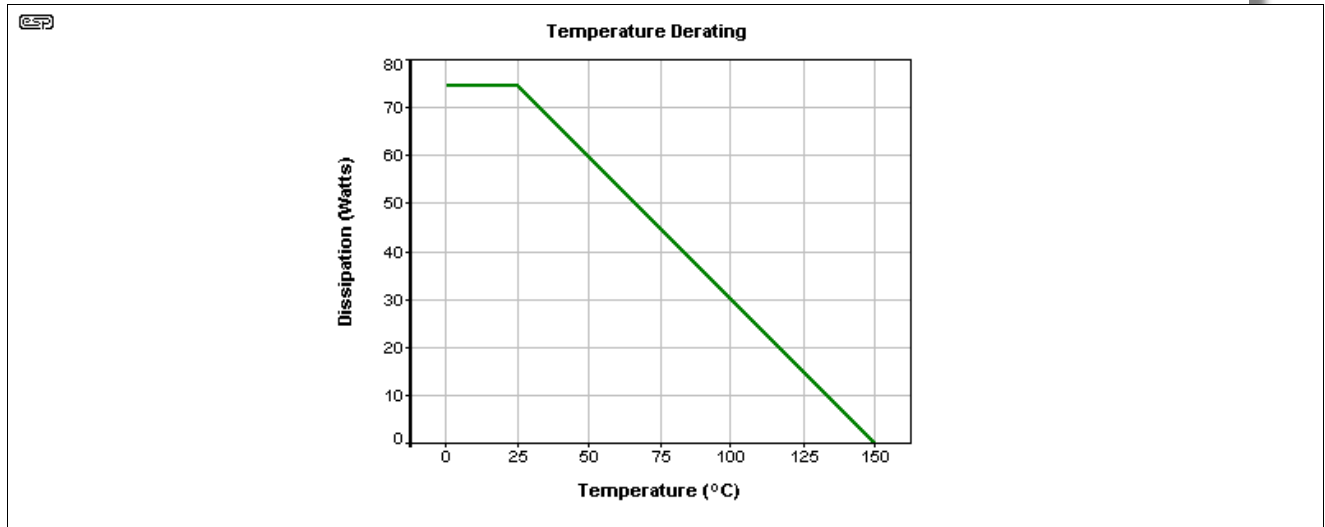




## 6 - Junction to Case Thermal Resistance

Although this is always an important consideration, there is nothing the amplifier designer can do to reduce this for any given transistor. Manufacturer's data will sometimes quote this figure, but it is more common to refer to the device case temperature - this is easier to deal with, since the data sheets have already taken due to case thermal resistance into consideration when the temperature derating graph was produced.

Derating is commonly applied to semiconductor devices, and typically the maximum power dissipation claimed for most devices will be at or below 25°C. At any temperature above this, less power is available from the transistor with increasing temperature, until at some figure (typically 150°C) the permissible power dissipation is zero. Modern SiC (silicon carbide) and GaN (gallium nitride) MOSFETs can operate at up to 175°C.



**Figure 3 - Case Temperature Derating For MJE3055**

Figure 3 shows the case temperature derating curve for an MJE3055 (TO-220) power transistor as an example. As can be seen, at 25°C and below, the device is rated at 75W dissipation, but at 100°C, 30W is the maximum permissible. At 150°C, no power may be dissipated at all.

This device has a 1.67°C thermal resistance from junction (die) to case (Motorola specification sheet for the MJE2955/3055), so the actual junction temperature will be somewhat higher than the case temperature until 150 degrees, at which no power may be dissipated, so case and junction will be at the same temperature.

Our first calculation shall be to see what the junction temperature will be at 75W device dissipation, with the case at 25°C:

Thermal resistance = 1.67°C/W

Power = 75W

Therefore junction temperature rise =  $1.67 \times 75 = 125^\circ\text{C}$

If the junction is 125°C above ambient (25°) then the total is  $125 + 25 = 150^\circ\text{C}$ . This is a seriously high temperature, and if you were to do the same calculation for all the temperatures on the scale above you will quickly see that the junction temperature is 150°C at maximum dissipation for any given case temperature. The thermal derating curve simply limits the allowed power dissipation to ensure that the junction never operates at above 150°.

The information above is obtainable from all device manufacturers, and is essential reading to ensure that transistors are not damaged by excess temperature. The maximum quoted temperature should never be exceeded, as the device will have a considerably lower life expectancy if overheated. Instantaneous failure is not uncommon if a device at an already elevated temperature is called upon to suddenly do some hard work.



You can work out the junction to case thermal resistance by dividing 125 by the power rating. A 125W power transistor has a 1°C/W junction to case thermal resistance.

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Using our theoretical amplifier above, we know that the dissipation will be 50W per transistor (or a total of 100W), and from the derating curve we see that for this power, the maximum case temperature is 65°C. The goal of this exercise is to determine the size of heatsink needed to ensure that the transistors ratings are not exceeded. We can also see if it is possible to operate them at a lower temperature, thus prolonging their operating life.

The junction to case thermal resistance varies widely, but will rarely be less than about 0.5°C/W, depending (of course) on the manufacturer, the case style, and the type of device. A few examples:

- IRF540 MOSFET - 1.0°C/W
- MJL21193/4 BJT - 0.7°C/W
- MJL3281A/1302A - 0.625°C/W
- MJE3055/2955 BJT - 1.67°C/W (our demonstration model)

Also remember that in use, the base-emitter voltage of transistors falls at -2mV / °C, and leakage current increases exponentially (leakage current doubles for every 8 to 10 °C increase in temperature). These are two very good reasons to try to keep the operating temperature as low as possible. The fall in base-emitter voltage requires that all Class-AB amplifiers have a 'bias servo' that compensates the bias voltage for temperature variations. Some power transistors have internal diodes (separately pinned) for the same purpose. These transistors have five leads (three for the transistor, and two for the internal diode).

## 7 - Case to Heatsink Thermal Resistance

This is where the whole process often falls down, since semiconductor devices are nearly always electrically insulated from the heatsink. This means that some material must be used between the case of the device and the heatsink surface. This is typically mica, ceramic (e.g. beryllium oxide or aluminium oxide) or Kapton - the latter is usually hard to get now for some reason, although it is available from ESP along with PCB purchases (it is *not* available separately). This will invariably increase the thermal resistance - there is no known material which is both a perfect electrical insulator and perfect thermal conductor (although diamond comes pretty close 😊). Sad but true. Heatsink compound ('thermal grease') *must* be used on both sides of mica, Kapton, aluminium oxide etc. to ensure minimum thermal resistance.

There is actually a number of alternatives for electrically insulating the transistor from the heatsink while still allowing heat transfer. Some were mentioned above, but there are others too. The list now includes oven bags - an unexpectedly good alternative ...

| Material                                | Thermal     | Electrical | Thermal Resistance | Other Properties                   |
|-----------------------------------------|-------------|------------|--------------------|------------------------------------|
| mica                                    | Good        | Excellent  | ~ 0.75 - 1.0       | Fragile (random thickness)         |
| Kapton (polyimide)                      | Good        | Excellent  | ~ 0.9 - 1.5        | Robust (but <i>very</i> thin)      |
| Oven Bag (high temp nylon) <sup>1</sup> | Very Good   | Excellent  | ~ 0.5 - 1.0        | Robust (but <i>extremely</i> thin) |
| aluminium oxide                         | Excellent   | Very Good  | ~ 0.4              | Fragile - easily damaged           |
| beryllia (beryllium oxide)              | Excellent   | Excellent  | ~ 0.25             | Dust is toxic                      |
| Sil-Pads                                | Fair +      | Excellent  | ~ 1.0 - 1.5        | Convenient (low power)             |
| phase-change                            | Good +      | Excellent  | ~ 0.8 - 1.5        | May be hard to find                |
| graphite                                | Outstanding | Conductive | ~ 0.2 - 0.4        | Delicate, no insulation            |
| Direct                                  | Outstanding | Conductive | ~ 0.1 - 0.2        | Must use thermal 'grease'          |

**Table 1 - Thermal Resistance of Various Mounting Methods (TO-220 Case)**

Note 1 While an oven bag (or part thereof) might seem unlikely (to put it mildly), this is an option I've wanted to test for some time, but never got around to it. These are usually made from a high temperature nylon, and while they won't survive a 'soldering iron test' (which is passed easily by Kapton), they are rated for a higher temperature than any semiconductor can handle. The typical thickness is around 10µm, versus 25µm for Kapton tape. Although (like most plastics) the nylon is a pretty horrible heat conductor, it works because it's so thin. I ran a test with a MOSFET on a heatsink, and I could not get the MOSFET's temperature to feel any hotter than the heatsink.

The table shows the approximate thermal resistance for various materials, and (except for the Sil-Pads and phase-change) assumes that a good quality thermal compound has been used. Despite the claims made by some manufacturers, most *good quality* thermal compounds are much of a muchness, since their primary purpose is to exclude air from the mating surfaces, and provide reasonable heat transfer



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themselves. Air is a most excellent thermal insulator, and even the tiny air gaps that exist between the transistor case, insulating washer and the heatsink will increase thermal resistance dramatically.

The next table shows the comparison between thermal *resistance* and thermal *conductivity*. Thermal resistance is dependent on the thickness of the material, and thermal conductivity is a quantitative measure of a material, expressed in watts per metre-Kelvin (W/m·K). In most cases it's not a useful way to describe a material, and thermal resistance is far more useful when designing a heatsink application. It's included here in the interests of completeness, but it's unlikely to be used by anyone. Thermal conductivity varies with temperature. Materials with a high thermal conductivity transfer heat more effectively than those with low thermal conductivity.

Note that thermal conductivity is a constant, but thermal resistance depends on the area and thickness of the material. The area is defined by the metal face of the transistor (or IC), *not* the physical size of the thermal material. As a simple example, a material with a thermal conductivity of unity (1W/(m·K)) will have a thermal resistance of about 0.25°C/W (or K/W) if the transistor pad is 10×10mm, and the material is 25µm thick. A larger surface area reduces the thermal resistance. There are on-line calculators you can use to make the conversion, but most seem to offer very odd material selections.

If the thickness of the TIM (thermal interface material) is doubled, so too is the thermal resistance. Note that this calculation is *only* for the TIM itself, and it doesn't include the inevitable additional thermal resistance between the transistor, TIM and heatsink. We minimise that by using thermal compound (aka 'grease'/'paste'), which will ideally be well loaded with microscopic thermally conductive particles. Ideally these fillers will be electrical insulators, such as aluminium oxide. Price is not necessarily an indicator of performance, but if you find a product that works well, stay with it. A common mistake is to assume that if a little is good, a lot must be better. This is *not* the case at all.

| Material                                | Thermal Resistance (°C/ W) | Thermal Conductivity (W/(m·K)) |
|-----------------------------------------|----------------------------|--------------------------------|
| mica                                    | ~ 0.75 - 1.0               | 0.71                           |
| Kapton (polyimide)                      | ~ 0.9 - 1.5                | 0.8                            |
| Oven Bag (high temp nylon) <sup>1</sup> | ~ 0.5 - 1.0                | 0.25                           |
| aluminium oxide                         | ~ 0.4                      | 30                             |
| beryllia (beryllium oxide)              | ~ 0.25                     | 209-330                        |
| Sil-Pads                                | ~ 1.0 - 1.5                | 1.1 (Claimed)                  |
| phase-change                            | ~ 0.8 - 1.5                | 1 - 8                          |
| graphite                                | ~ 0.2 - 0.4                | 168                            |
| Aluminium                               | N/A                        | 237 (Typical)                  |
| Copper                                  | N/A                        | 401                            |
| Steel                                   | N/A                        | 43 (1% Carbon)                 |
| Diamond                                 | N/A                        | 1,000 (Reference)              |

**Table 1A - Thermal Resistance Vs. Thermal Conductivity of Mounting Materials & Metals**

Heatsink compound (aka 'thermal grease' or 'TIM' - thermal interface material) is a purpose-made product - never use ordinary silicone grease or anything that is not specifically designated as thermal compound. Most heatsink compound uses ultra-fine suspended particles of zinc oxide, aluminium oxide or other thermally conductive but electrically insulating material. Make sure that the compound you get is electrically non-conductive! (Unless using direct mounting, where it doesn't matter either way.) The figures for thermal resistance are based on the average thickness, which varies depending on the material. Thinner materials have lower thermal resistance, but are more easily damaged, potentially allowing a short from the transistor case to the heatsink.

**Mica** is (was) probably the most commonly used insulator, but it has limitations. The worst of these is quality control, which results in mica washers being whatever thickness they happen to be (i.e. unpredictable). With infinite care and patience it is possible to shave mica down to a thickness of perhaps 0.05mm (0.002"), but few of us have the time to fiddle about at this level. A micrometer is essential if you really want to do this. See the section on mica (below) for more information on this topic.

**Kapton** is an excellent material in this respect. Predictable thickness and very tough (try ripping it), but generally not quite as good as mica thermally. For reasons which remain entirely obscure, Kapton transistor washers do not seem to be as readily available as they once were, which I find a real shame since they have always been a favourite of mine. However, Kapton tape (as noted above, available from ESP) can be made into very acceptable washers with a pair of scissors and a hole punch.



Absolute cleanliness is essential, since the tape is typically only 25µm (0.001") thick, and the smallest piece of swarf will puncture it. To put this into perspective, a human hair is between 17 and 180µm in diameter - with an average of around 70µm.

**Oven bag** material is very good - I figured it would be good, but it's better than I expected. These used to be very common in the kitchen, and while they have fallen from favour they are still available (and are quite inexpensive - a pack will provide hundreds of insulators). Because it's so thin, it's subject to puncture by the tiniest bit of conductive material (e.g. swarf), so absolute cleanliness is essential. The oven bag I tested is 10µm thick, less than half that of Kapton. It's easy to cut with scissors, and you don't need to make a 'proper' hole - just puncture it where the screw hole is, and push the screw through (I generally use the same technique with Kapton). Despite the extremely thin material, I went for broke and tested the insulation resistance with a 1,000V insulation tester - nothing! There was no sign of breakdown, even though the test voltage is typically ten times that encountered in normal use. Under no circumstances would I recommend it for isolating mains voltages though. It will not survive a soldering iron test - it starts to melt (unlike Kapton, which can withstand a soldering iron until you get bored). However, it is rated to handle cooking temperatures, so can handle any temperature a semiconductor can provide without spontaneous destruction. *Note that if you use this material, you do so entirely at your own risk.*

**Aluminium oxide**, in the form of hard anodised aluminium, is an excellent electrical insulator, and provides very good heat transfer, but is fragile and easily damaged. Even a small scratch will generally allow an electrical short circuit, and obtaining a consistently thick (relatively speaking) layer of oxide is difficult in the extreme in mass produced heatsink extrusions. Hard anodised aluminium washers are available from some electronics suppliers, but are generally expensive and you will probably find that they are not available for all transistor case styles. An alternative is aluminium oxide ceramic - essentially just what it sounds like. Al<sub>2</sub>O<sub>3</sub> (aka alumina) ceramic looks very much like beryllium oxide ceramic, and although not as good thermally, is still far better than mica, Kapton, etc. Alumina washers are available in a variety of shapes to suit most transistor case styles from many of the larger parts suppliers.

**Beryllium oxide** was quite popular a few years ago, until someone discovered that it is toxic. Consequently, such mounting washers have all but vanished (at least as far as the DIY fraternity is concerned). A pity, since most people I know have never eaten a transistor washer in their lives, but governments do so like to protect us from ourselves. In fact, it is *inhalation* of the beryllium oxide *dust* that causes problems (respiratory disease) rather than ingestion, and it turns out that they are still available, as are many other parts made from this unique and versatile ceramic. Because it *is* toxic, special precautions are needed during manufacture, and this makes it rather expensive and difficult to obtain.

**Silicone insulators** (Sil-Pads™ or similar) are only comparatively recent additions to the mounting hardware options, and are very convenient to use. The down side is that they are nowhere near as good (thermally) as thin mica or Kapton, but they require no silicone grease or other thermal compound. Re-using silicone pads is unwise, because the deformations caused by prolonged heat and pressure will cause even greater thermal resistance if they are re-used.

I have tested and verified this, and it makes perfect sense, since the thickness of the pad will not be even after it has been used, due to uneven pressure and mating surfaces (at a microscopic level). Even if the new device is located in exactly the same place on the pad, thermal resistance will be impaired. I suggest that Sil-Pads be used where maximum thermal conductivity is relatively unimportant, as their thermal resistance is generally not as low as claimed. The figure shown in Table 2 for the TO-3 and TOP-3 case styles is (IMO) highly optimistic, and is unlikely to be achieved in practice. In general, I suggest that silicone rubber pads are used only where devices operate at low power, regardless of manufacturer and claims. I have *never* used silicone pads of any type at high power and been happy with the result. In every case where I thought I might get away with it, I've had to remove the silicone and revert to Kapton and heatsink compound.

**Phase-change** materials are relatively recent, and I have not had the opportunity to run any tests. They may be difficult to find, and appear to be comparatively expensive from the little info I've been able to locate so far. They appear to be based on wax, loaded with thermally conductive material. The phase-change (from solid to semi-liquid) occurs at around 60°C, and allows the material to 'flow' into any voids (air gaps) between the transistor and heatsink. Some are available with an insulating film (typically Kapton), and others are applied via an applicator or in sheet form with a 'carrier' or backing material for transport.



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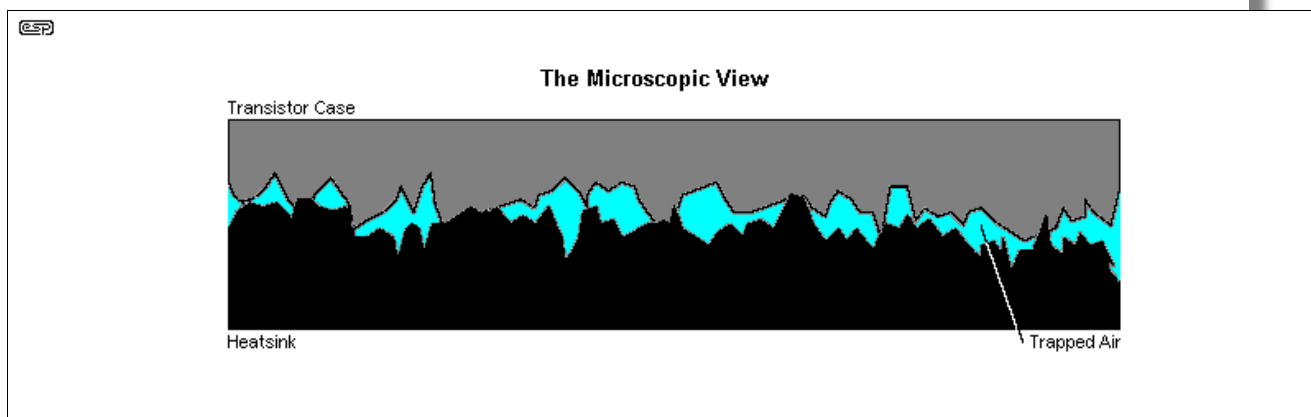
**Graphite** is an outstanding thermal conductor, but unfortunately it's also highly electrically conductive. This means it can only be used where the transistor's mounting face (typically the collector or drain) doesn't need to be insulated from the heatsink. This may be because the heatsink is 'floating' at some voltage, or that the collector or drain is at earth/ ground potential and insulation is not needed. It's almost never used in audio (I've certainly not seen it), because there are nearly always multiple devices on the heatsink and they must all be electrically insulated from the heatsink and each other. I have seen it used in switchmode power supplies where a live heatsink is utilised (these can be very dangerous of course).

| Package       | Direct (Dry) | Direct + Grease | Mica + Grease | Sil-Pad™ |
|---------------|--------------|-----------------|---------------|----------|
| TO-3          | 0.5 - 0.7    | 0.3 - 0.5       | 0.4 - 0.6     | 0.4      |
| TO-264        | 0.5 - 0.7    | 0.3 - 0.5       | 0.4 - 0.6     | 0.5      |
| TOP-3/ TO-218 | 0.8 - 1.1    | 0.5 - 0.7       | 0.6 - 0.9     | 0.65     |
| TO-220        | 1.1 - 1.3    | 0.9 - 1.1       | 1.0 - 1.6     | 1.5      |
| TO-126        | 1.5 - 2.0    | 0.9 - 1.2       | 1.2 - 1.7     | NA       |

**Table 2 - Comparison of Case Types**

The information in Table 2 shows the variations expected with various transistor cases and mounting techniques ('Grease' refers to heatsink compound). It should come as no surprise that these figures are different from those in Table 1, since this info is from a different source. I prefer to err on the side of caution when it comes to thermal resistance, because it is always better to have the devices run a little bit too cool than a lot too hot. I would suggest that Table 1 is more realistic (if not all that accurate).

As mentioned (briefly) above, you can work out the junction to case thermal resistance ( $R_{th(j-c)}$ ) for any device simply by dividing 125 by the 25°C power rating. The maximum dissipation is always quoted for a case temperature of 25°C, and the maximum junction temperature is nearly always 150°C (some MOSFETs can operate at higher temperatures). An example 200W transistor will (by calculation) have a junction to case thermal resistance of 0.625°C/W. Using this technique will work every time, and is probably the easiest way to get the figure if it's not stated in the datasheet. A device rated for 125W has a junction to case thermal resistance of 1°C/W.



The importance of a thermal compound cannot be overstated. The drawing shows the surface of the transistor and heatsink (or mica washer) at a microscopic level. OK, so it's just a drawing, but this is exactly what you would see under a powerful microscope. The blue area represents air, and the surfaces only touch in a few places. This is why the thermal resistance is so much higher without the use of any thermal compound. The thermal compound (or other interface material) fills the gaps to exclude air and provide a greater overall contact area.

In some cases the situation is worse than the drawing - I wonder how many readers have come across (especially) TO-220 transistors that look as if they had been machined with a chainsaw. A flat file, or fine wet/dry abrasive paper and water (no, it won't hurt the transistor) should be used to make sure that the surface is as flat and smooth as you can make it. In many instances, a device with poor finishing of the metal face may be indicative of a fake - counterfeit transistors are almost guaranteed if you buy at a very low price from any online auction site.

**Never hold the transistor in a vice and file it!** Hold the file still, and gently slide the transistor on the file (or wet/dry paper on a sheet of glass) until it is smooth. There is no point having it nice and smooth if it is rounded - this *will* happen if you rigidly mount the transistor and hand-hold the file.



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When you are finished, you should be able to lay the transistor on the heatsink and see no light between the two surfaces, from any observation angle. Use a bright light behind the heatsink so you can see any surface imperfections. This little bit of extra effort may mean the difference between the success or failure of your project - at least in the long term.

Direct mounting (with thermal 'grease') is recommended where either the collector (or drain for a MOSFET) is at ground potential, or if you can insulate the heatsink from the chassis. The latter is unwise (in the extreme) for any external heatsink, but works better than any other method of mounting if the heatsink is internal. It's almost always impractical for power amplifiers, but for other applications where maximum heat transfer is needed, you won't get better. Using a 'hot' heatsink (i.e. at some voltage above/ below the earth/ chassis potential, AC or DC) is not for the faint hearted, and it should have a clear warning sticker attached as a warning to others who may service the equipment later.

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## 8 - Mica

What is Mica? (from Wikipedia) - The mica group of sheet silicate (phyllosilicate) minerals includes several closely related materials having close to perfect basal cleavage. All are monoclinic, with a tendency towards pseudo-hexagonal crystals, and are similar in chemical composition. The nearly perfect cleavage, which is the most prominent characteristic of mica, is explained by the hexagonal sheet-like arrangement of its atoms.

The word 'mica' is derived from the Latin word mica, meaning 'a crumb', and probably influenced by micare, 'to glitter'.

Mica is transparent, and can be split into a very thin film along its cleavage. Electrically, it has the unique combination of high dielectric strength, endurance, uniform dielectric constant, low dielectric loss (or high Q), and extremely good insulating properties. Mica is moisture-proof and has low heat conductivity. It is infusible and may be exposed to high temperatures (in excess of 700°C) without any noticeable effect. If you want more info - try a web search 😊.

So much for the descriptions, but why would mica get a section all of its own? Simply because the quality control is virtually non-existent (from what I have seen lately). The overall shape is fine, but the thickness is generally too great - commonly by a huge margin. I have used mica washers that were so thick that I was able to separate them with a (very) sharp knife, and obtained four washers for the price of one (plus some scrap from the splitting operation). Because mica has low heat conductivity, it has to be as thin as possible to ensure acceptable thermal resistance.

Splitting a mica washer is not hard, but requires a steady hand, a good magnifier, and the sharpest scalpel you can find (or a new razor blade). The thickness should ideally be in the order of 0.025 to 0.05 mm (25 - 50µm, or 1 - 2 mil) for normal use, but you will probably find that up to 0.1 mm is acceptable for low power devices. It is possible to make it much thinner than this, which decreases thermal resistance but makes the washer fragile and easily damaged.

The electrical characteristics of mica are such that even the thinnest possible washer will be more than adequate for typical amplifier voltages. The dielectric strength allows mica to withstand 1,000 - 1,500 volts per mil (1/1,000 inch), or about 0.025 mm (25 micrometres) of thickness without puncturing or arcing.

This being the case, I am unsure why commercially available transistor washers are anything up to 0.25mm thick (and no, I am not kidding - I have even found some thicker than that!). This is capable of withstanding over 10kV in theory, let alone a measly 200V or so, and will introduce considerable thermal resistance. If one had a steady enough hand, just one of these would yield 10 washers of acceptable thickness (thinness?), with each still capable of at least 1kV insulation breakdown.

I am even tempted to make a mica splitter, that can be adjusted to a suitable thickness. One of these days ... (or not - I use Kapton almost exclusively).

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## 9 - Thermal Compounds

One of the most common mistakes made by hobby electronics enthusiasts (and quite a few professionals too), is to assume that if a little thermal compound is good, a lot must be better. Absolutely not so! The amount of thermal compound should be exactly that amount which ensures that an air-free join is made between the mating surfaces. If too much is applied it will cause an increase in



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thermal resistance, since it is not really that good at conducting heat. Generally speaking, any electrical insulator is also a thermal insulator (there are a couple of exceptions), so the thinner the final composite insulation - including thermal 'grease' - the better.

Having said that, one must ensure that the electrical insulation is sufficient for the applied voltage or disaster will surely follow - usually in a spectacular fashion - especially if high voltages and currents are available.

Although several manufacturers over the years have thought they could get away with using silicone grease with no fillers, don't! It doesn't work, and eventually flows out from under the transistor leaving the thermal connection dry and causing device overheating and failure. Always use a good quality thermal compound, and make absolutely certain that it is non-conductive if the transistors are to be insulated from the heatsink. Some of the specialised compounds for CPU cooling in PCs are electrically conductive, and cannot be used where electrical insulation is needed.

Because you use so little thermal compound, I suggest that you invest in good quality. Some are marginal - they're cheap, but you may find out why after a few years when output devices fail. Both the filler (usually an oxide of some kind) and carrier 'grease' (almost always silicone) must be appropriate for the purpose. You need plenty of filler, and a carrier that is sufficiently viscous to keep the filler in suspension, but is soft enough to allow you to get a thin, even coating.

A quick note on the application of thermal compounds is in order. This is the method I generally use, and it works very well once you have a copious supply of workshop rags or paper towels to clean the mess from your fingers.

Apply a small quantity of the thermal compound to one finger, then gently rub with your thumb to create an approximately even coating on finger and thumb. Now rub the thermal compound onto a washer, held between thumb and finger, ensuring that the coating is just thick enough to be opaque, but thin (and even) enough to ensure that the contact will be absolute on both surfaces (transistor and heatsink). Test your skill until with moderate pressure, you can leave a perfectly formed outline of the transistor, with no blobs or gaps, on the heatsink surface.

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In some cases, you may find that there isn't a great deal of heat to remove, but doing so is somewhere between difficult and impossible. In reality, probably not. There are several thermally conductive potting compounds available, and one that has been recommended to me comes from ITW (Insulcast) in the US. There are others from many suppliers, but some are harder to work with than others. The data sheets will usually provide information that can help you make a selection. Don't expect especially high thermal conductivity - these compounds are only suitable for low power devices.

Where such a compound is appropriate, you will normally need something that has a very low viscosity, so it will flow easily into narrow gaps between the semiconductors and the heatsink. The heatsink (or case) itself will need to be heavily anodised or otherwise insulated so that semiconductor cases cannot short circuit to the case while the unit is being assembled.

If such a compound is needed, be prepared for at least some frustration finding a suitable material and a local supplier.

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## 10 - Thermal Inertia

Since any heatsink has some mass, it will also possess thermal inertia, which is to say that it takes time for the body of heatsink material to heat up. Naturally, the larger the heatsink (and the more aluminium it has in it), the longer it will take to heat. This can easily lead one to believe that the heatsink is large enough for the job. Only by running an amp for an extended period will the reality reveal itself - especially if there is not enough surface area to allow the heat to dissipate into the atmosphere.

Thermal inertia is a good thing, because it allows the heatsink to absorb 'transient power surges', and will dissipate the heat again during low power operation. Because of the dynamics of music, this tends to work well, but for professional use (where heavy compression is often used), the average power into the heatsink can be very high for prolonged periods, so either massive bulk material must be used, or a combination of reasonable bulk and lots of surface area. Fan cooling is almost mandatory for professional use at high continuous power levels.



Even if one were to obtain an infinitely large block of aluminium, if a bracket or other mounting arrangement directly underneath the heat source (the transistors) is not thick enough, it will have significant thermal resistance, and the transistors may just overheat anyway, so we do need to look at all the resistances in the thermal circuit, not just the heatsink itself.

It is not uncommon to have transistors operating at well in excess of their thermal ratings, but a casual 'finger' test of heatsink surface temperature will appear to indicate that all is well.

Thermal inertia can be stated as specific heat - the amount of energy (in Joules) needed to raise the temperature of one gram of a material by 1°C (or 1 Kelvin). For the same mass, aluminium is a much better choice than copper - assuming of course that you were planning a copper heatsink. For a given weight of material, aluminium requires over twice as much heat input as copper to raise its temperature by 1K (1°C). This means that an aluminium heatsink will absorb more energy for the same temperature rise, providing a useful thermal buffer. Then all we have to do is dissipate that heat energy into the air as quickly as possible.

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## 11 - Minimising Thermal Resistance

The key to obtaining maximum power from any transistor is minimising the thermal resistance from junction to free air, and as can be seen from the above information, one of the worst offenders is the transistor itself. Most aspects of the transistor's thermal characteristics (whether bipolar or MOSFET) are determined by the manufacturer, and it would seem that the user has little control. This isn't necessarily the case ...

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### 11.1 - Mounting The Transistors

The most common way to mount the transistors is to use metal thread screws, either with nuts on the back of the heatsink, or with a thread tapped into the heatsink itself. In production environments, spring clips are often used, and even rivets are sometimes used for mounting. The latter is a very bad idea, as the following tale describes ...

Transistors, especially the TO-220 are sensitive to mechanical shocks. The TO-220 for example should never be attached to its heatsink by means of a rivet, either conventional or pop rivet. I was building replacement voltage regulators for a large automotive manufacturer, where the original device was riveted to a metal base with a mica insulator in between. Our version of this worked okay for a while until one day the manufacturing setup guy over adjusted the rivet force and there was some deformation of the tab. The parts kept passing the 100% electrical test and the manufacturing group kept making them over my objections. This too worked okay for awhile until we started getting field failures. When we investigated, we found that the power Darlington's inside were suffering 'earthquake' damage due primarily to the shock of the rivet machine. Often it was the outer edges of the device that would crack, leaving the device functioning but affecting  $V_{ce\ Sat}$ . With the higher saturation voltages (not measured on our tester at the time) the devices overheated and failed.

It took 7 tractor trailer loads to haul all of the inventory back from our warehouse. We had to rework over 100K voltage regulators.

Having seen the problem once, you would be amazed at how often I have seen it since. One guy was using a pop rivet gun. There was no damage to the tab but the shock from the pop of the rivet sent its message through the tab to the silicon. I saw another guy who was cutting the tab off the device in order to package it in the available space. I saw a piece of home brew equipment where the tab was being held by a nut and bolt that had been over tightened distorting the tab. Your comment about not holding the device in a vice and not filing on it with a hand file basically speak to the same issue: Silicon is thin and brittle. Do not alter the package or subject it to shock

My thanks to Mike for that story. I have seen transistors riveted to heatsinks as well, but that was many years ago, and I don't know if it caused any problems. The main issue is that riveting is at best unpredictable, and also creates considerable shock when the mandrel breaks (for pop rivets at least).

The key to mounting is pressure - it must be just right. Too much, and you distort the device case and possibly damage the thread or cause protrusions in the soft aluminium heatsink. Too little, and the thermal resistance will be too high. High reliability applications (Mil-Spec) will demand that a torque driver is used, to ensure that every device is tightened to exactly the right pressure. I don't suggest for an instant that this is needed in your project (torque drivers tend to be rather expensive), but it is essential that you develop a feel for tightening the screws to get consistent and reliable results.



| Package                        | Screw Tightening Torque               |
|--------------------------------|---------------------------------------|
| TO-220                         | 0.490 to 0.686 N · m (5 to 7kgf · cm) |
| TO-220 Full Mould              | 0.490 to 0.686 N · m (5 to 7kgf · cm) |
| TO-3P                          | 0.686 to 0.822 N · m (7 to 9kgf · cm) |
| TO-3P Full Mould               | 0.686 to 0.822 N · m (7 to 9kgf · cm) |
| TO-3P two-point mount (Sanken) | 0.686 to 0.822 N · m (7 to 9kgf · cm) |

Table 2A - Transistor Mounting Torque [8]

Once the transistor mounting technique is perfected, we can generally assume that the thermal resistance will be about  $1^{\circ}\text{C}/\text{W}$ . This may be bettered, but again, a safety margin is always useful. At a continuous dissipation of 50W, this means a temperature rise of  $50^{\circ}\text{C}$  due to mounting and insulation requirements. Since we already worked out from the derating graph that the absolute maximum case temperature is  $65^{\circ}\text{C}$  for 50W dissipation, we are faced with a problem - the maximum heatsink temperature is only  $15^{\circ}\text{C}$ . If this were to be the (somewhat chilly) ambient temperature, a  $0^{\circ}\text{C}/\text{W}$  heatsink is needed!

Perhaps I have been looking in the wrong places, but I have not been able to find a heatsink which can manage  $0^{\circ}\text{C}/\text{W}$ , and if the ambient temperature were to rise to (say)  $20^{\circ}\text{C}$ , the heatsink now needs a negative thermal resistance - this is called a heat pump, and the most common example is an air conditioner or refrigerator. This approach, although quite feasible, can become somewhat expensive. Peltier devices (hot/cold junction semiconductors) can be used, but tend to be expensive, and require a fairly heavy current. In order for the cold junction to work at a suitably low temperature, the maximum temperature of the hot junction must be kept as low as possible - this requires (only one guess!) ... a heatsink.

For the above example, because there must always be a thermal gradient (temperature difference) to allow heat to flow from one device to another, this means that ambient temperature must be  $-10^{\circ}\text{C}$  to be able to use a  $0.5^{\circ}\text{C}/\text{W}$  heatsink! We have just designed an impossible output stage, which cannot be made to work in real life without considerable additional cost.

## 11.2 - Further Reductions of Thermal Resistance

It is generally far cheaper to reduce the thermal resistance of the devices and their mountings than it is to use the largest heatsink you can possibly find, or go to all the bother of water cooling (which works extremely well, but is difficult to implement in the listening room), or using fans which are noisy and spoil the signal to noise ratio of your equipment. There is no point ensuring that the S/N ratio is 80dB or more, only to have a background noise of perhaps 45dB SPL created by cooling fans. Remember that even if the heatsink stays at ambient temperature (an infinite amount of air will be needed 😊) from the above example, the absolute maximum ambient temperature is  $15^{\circ}\text{C}$ .

One technique used to reduce thermal resistance is simple - use two (or more) transistors in parallel in place of a single device. Although the thermal resistances for each of the transistors remain the same, the resultant thermal resistances for a 'parallel pair' are effectively halved. This is due to the fact that each transistor is only dissipating half the total power.

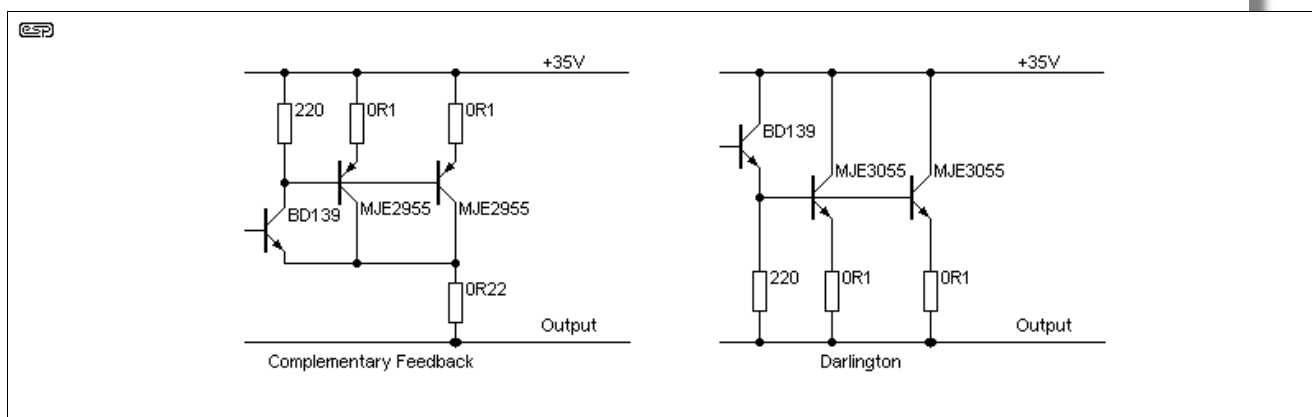
Heatsink calculations for the parallel pair are best carried out by considering each transistor individually. In this example, a single transistor is expected to dissipate 50W, so each transistor of a parallel pair has a power dissipation of 25W.

From Figure 3, the maximum case temperature must not exceed  $105^{\circ}\text{C}$ . With a case to heatsink thermal resistance of  $1.0^{\circ}\text{C}/\text{W}$ , there will be a  $25^{\circ}\text{C}$  temperature difference between the two. The maximum heatsink temperature is therefore  $80^{\circ}\text{C}$  (at this temperature, protection would need to be provided to prevent accidental contact, but this *must not* affect airflow across the heatsink). If we assume an ambient temperature of  $30^{\circ}\text{C}$ , each transistor requires a heatsink with thermal rating of  $2^{\circ}\text{C}/\text{W}$ . The parallel pair will require a heatsink of twice this size, i.e.  $1.0^{\circ}\text{C}/\text{W}$ . This calculation does not allow for any safety margin (other than that built-in to the case to heatsink thermal resistance assumptions) and it would be better to design using a higher ambient temperature.

For an ambient temperature of  $40^{\circ}\text{C}$ , the heatsink for the paralleled pair would need to have a rating of  $0.8^{\circ}\text{C}/\text{W}$  and for an ambient of  $50^{\circ}\text{C}$ ,  $0.6^{\circ}\text{C}/\text{W}$ . If we are going to mount all the transistors for a single amp on the one heatsink (which is the most common approach), then the heatsink must have a rating of  $0.5^{\circ}\text{C}/\text{W}$  for an ambient temperature of  $30^{\circ}\text{C}$  and no safety margin. The ratings at  $40^{\circ}\text{C}$  and  $50^{\circ}\text{C}$



would be  $0.4^{\circ}\text{C/W}$  and  $0.3^{\circ}\text{C/W}$  respectively. Similar calculations can be carried out for any number of paralleled transistors.



**Figure 4 - Connecting Transistors in Parallel**

The 0.1 Ohm resistors shown in Figure 4 ensure that each transistor carries the same (or at least roughly equal) current. Without them, the transistor with the higher gain (or the lower emitter-base voltage  $V_{be}$ ) will take the majority of the current. This will cause it to get hotter than the 'lesser' transistor, which in turn increases its gain further and lowers  $V_{be}$  even further, which means it will get hotter, and so on. Note that the use of 0.1 ohm resistors is restricted to transistors with fairly closely matched  $V_{be}$ . Higher values are recommended where matching is impractical. This is all standard stuff in amplifier design. The 0.22 ohm resistor is used to stabilise bias current and introduce local feedback. This must not be omitted.

There are other ways in which thermal resistance can be reduced. For example, we are assuming that the transistors will all be on the same heatsink, which will probably be earthed to the chassis. If the NPN and PNP transistors were to be mounted on separate heatsinks which were not earthed but connected to the power supply (or output - depending on the topology of the output stage), we can reduce the case to heatsink thermal resistance even more.

This approach is generally considered too much of a nuisance though, because of the risk with high supply voltages, and the difficulty of mounting the heatsinks using insulating bushes, nylon screws etc. Also, it will be necessary to provide some form of shield, such as perforated steel mesh, to protect the heatsinks from becoming shorted to each other or the chassis should some object be dropped. Even the end of a lead swinging about could destroy the amplifier - not a happy thought.

There is also a topology where the transistors are bolted directly to an earthed heatsink, but the power supply is allowed to float - connected to the amp's output. While this technique certainly reduces the case to heatsink thermal resistance, it makes the power supply harder to design and introduces other complications.



For a typical Class-AB audio amplifier, the 'real world' outlook is usually not as pessimistic as implied above. This is simply because the dissipation is not continuous, but varies with the music and power output. Therefore, it is not necessary to design for continuous worst-case dissipation, but for some lower long-term average. For the output stage discussed here, a realistic heatsink rating is around  $1^{\circ}\text{C/W}$  for a nominal 70W Class-AB amplifier used for hi-fi (with  $\pm 35\text{V}$  supplies). If pushed into service with dance music (for example), the average power will often approach the worst-case and a larger heatsink is called for.

It's worth noting that the old FTC (US Federal Trade Commission) test included 'pre-conditioning' the amp before power tests, with the amp operated at  $\frac{1}{2}$  power for one hour (this has been changed to  $\frac{1}{8}$  power). The original test was *brutal*, as it was right at the power level that caused maximum dissipation in the output transistors, and was completely unrealistic compared to how most amplifiers are used. The peak to average ratio is typically around 10dB for 'modern' music, so a 100W amp will have an average output power of 10W when driven to the onset of clipping.





For reasons that I find somewhat depressing, some manufacturers/ sellers of thermal pads (in particular) show the thermal *conductivity*, rather than thermal resistance. You may well ask why this is depressing, and the answer is simple ... it prevents people from being able to make direct comparisons (either that or it's just simple bastardry). Thermal conductivity is measured in  $W/(m \cdot K)$ , being watts per metre-Kelvin (1K is equal to  $1^{\circ}C$ ). There's a table in [Section 15.1](#) showing the thermal conductivity for a number of materials, and it now includes a number of thermal interface materials.

Fortunately, there's a simple way to convert between the two measurements, as shown below. If faced with this quandary is to look for the material with the *highest* thermal conductivity (largest number) when comparing materials. I obviously can't be sure, but I suspect that this technique is used to confuse the purchaser by failing to show just how bad the interface material really is. You need to know the area of the transistor's metal face. For example, a  $17mm \times 19mm$  metal face (TO-264/ TO-3P) has an area of  $323\mu m^2$  (323 square mm).

To convert from thermal conductivity ( $W/(m \cdot K)$ ) to thermal resistance ( $K/W$  or  $^{\circ}C/W$ ) use the formula ...

$$R_{th} = t / k \times A \quad \text{where ...}$$

t is thickness in metres  
k is thermal conductivity in  $W/(m \cdot K)$   
a is area in square metres

So, when you see a thermal interface with a thermal conductivity of (say)  $1.2W/(m \cdot K)$  and it's 0.5mm thick, you should understand that it's dreadful. Consider that [concrete](#) (yes, actual concrete) has a thermal conductivity of around  $1.0-1.8W/(m \cdot K)$ , it's pretty obvious that the advertised thermal interface material is either *worse than* or not much better than a very thin piece of concrete, and is completely unsuitable for anything that dissipates more than a (very) few watts.

Not surprisingly, this method of describing the 'product' is primarily used with silicone pads, sheets or rolls. While they are useful for low-power devices, they are completely *unsuitable* for anything that will dissipate more than a few watts. I don't recommend them in any project where the dissipation is expected to be more than 10W, and while they are useful in low-power applications I've never liked them much, and any claim that they are suitable for high-power devices is blatantly false. There are some materials that appear very good indeed, but you quickly discover that they are very soft and are not intended for electrical isolation of more than a few volts.

In some cases, you may be able to find published information that provides both thermal conductivity *and* thermal resistance/ impedance. Unless the silicone is very thin (no more than 0.2mm thick which is very uncommon), the thermal resistance will always be very disappointing. One I looked at claims  $1.8W/(m \cdot K)$ , and at 0.008" thick (0.203mm) it has a thermal resistance of  $0.3^{\circ}C/W$  (TO-264 case). While that's actually pretty good, it's an outlier - all the others in the same brochure were worse! Many were over  $3^{\circ}C/W$  - a mere 10W will raise the semiconductor case temperature by  $30^{\circ}C$ . (No, I won't provide a link to the brochure, as I have no intention of 'advertising by proxy' by including one.)

#### 11.4 - Additional Thoughts On Reducing Thermal Resistance

Some other methods also exist for reducing the thermal resistance, with one of the simplest being to use a higher power transistor, or even a simple change of case type. Use a TO-3 case or large footprint plastic package instead of a TO-220 package for example. Here a few real examples ...

| TO-3 package          | Watts | Max Temp.       | Thermal Resistance                       |
|-----------------------|-------|-----------------|------------------------------------------|
| 2N3055                | 115W  | 200 $^{\circ}C$ | 1.5 $^{\circ}C/W$                        |
| MJ802                 | 200W  | 200 $^{\circ}C$ | 0.875 $^{\circ}C/W$                      |
| TO-3P/ TO-264 package | Watts | Max Temp.       | Thermal Resistance                       |
| TIP3055 (TO-220)      | 90W   | 150 $^{\circ}C$ | 1.39 $^{\circ}C/W$ (16 x 20 mm)          |
| TIP35                 | 125W  | 150 $^{\circ}C$ | 1.00 $^{\circ}C/W$ (16 x 20 mm)          |
| 2SC5200               | 150W  | 150 $^{\circ}C$ | 0.83 $^{\circ}C/W$ (20 x 26 mm) (TO-264) |
| MJL21193              | 200W  | 150 $^{\circ}C$ | 0.70 $^{\circ}C/W$ (20 x 26 mm) (TO-264) |

Table 3 - Transistor Case Styles

The figures quoted above are typical, and will vary (sometimes significantly) depending on the manufacturer and the fabrication method used.

As you can see, simply selecting a higher power device in the same case style, the thermal resistance



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is lowered, since the manufacturer must make allowance for the junction to case thermal resistance to allow the higher power. In many instances, this will probably be 'automatic', since the silicon die must be larger to allow more power, and thus has greater surface area and better heat transfer.

## 12 - Introduced Thermal Resistances

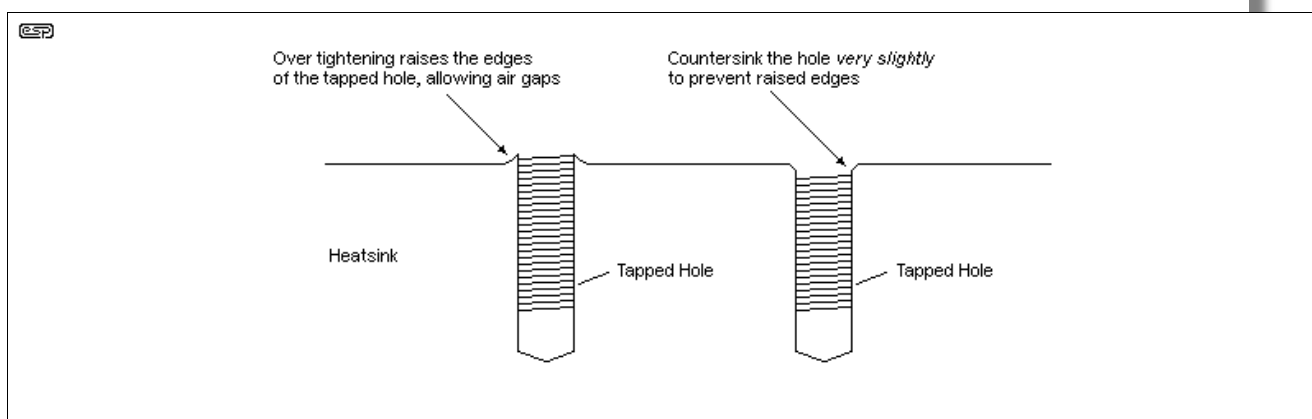
It is not uncommon for amplifier designs to use a bracket of some sort to connect the transistors to the heatsink. Where this is done, it is imperative that the minimum additional thermal resistance possible is created. I have seen designs where a simple aluminium bracket is simply bolted onto the heatsink, and everyone hopes that this will be enough. Generally, it is not. Transistors must be mounted as close as possible to the heatsink side of the bracket, since the aluminium is not a perfect conductor, and the smaller the distance the heat has to travel the better.

Any bracket used must be of the thickest material possible, and be mated to the heatsink with the greatest care. Otherwise, one is simply adding an unknown (possibly quite large) thermal resistance between the transistor's junction and the ambient air. I have heard, and seen in magazine readers' letters, complaints from amplifier builders that the transistors keep blowing for 'no reason'. Other more experienced (or careful) constructors have no such problems with the same design, and it is quite rare that the designers of magazine construction projects will suggest that the bracket to heatsink mounting could well be the problem.

If brackets must be used, ensure that the mating surface with the heatsink is completely flat, and free of burrs or other anomalies which will prevent the 'perfect' contact. There should be no less than 1 screw (or aluminium rivet) for each square 25mm of surface area, and screws should be tightened from the centre outwards, much as one tightens the cylinder head bolts on a car engine. A thin film of thermal compound is a must, and the mating should be tested with firm pressure then inspection to ensure that the contact surface is even. If it is not, then fine valve grinding paste (available from automotive parts stores) can be used to ensure that the surfaces are matched to each other.

Simply place a small quantity of grinding paste on the bracket, and distribute it with your finger. Rub the two components together (bracket and heatsink) in small circular patterns, until inspection shows that the two surfaces are evenly polished. It will not hurt one little bit to use a metal polish to shine the surfaces when you are satisfied that they mate properly - the smoother the surfaces, the better the thermal contact, since surface irregularities are minimised. Thermal compound should be used sparingly - once the surfaces are mated, the smallest amount only should provide coverage of the entire surface.

When properly lapped together, with only thermal paste (or perhaps some oil) it should be almost impossible to pull the surfaces apart. Air pressure should keep them mated, and you should have to slide them apart before final cleaning and assembly.



**Figure 4a - Raised Edges Due To Aluminium 'Stretching' & The Solution**

A short word about the mounting holes and screws is also called for. When a screw is tightened into a soft metal such as aluminium, it will tend to distort (or stretch) where the screw exerts pressure. On the heatsink, this will cause the metal to distort outwards, preventing the proper mating of the surfaces. After all your hard work ensuring that the surfaces were as good as you could get them, this will instantly negate your efforts. *Small* recesses in the underside of the bracket's screw holes will allow the distortion to occur, but will not lift the bracket off the heatsink. Likewise, a *very slight* recess in the

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surface of the heatsink will prevent the distortion from projecting above the surface - I tend to use both methods at once just to be sure. It is vitally important that any recess or countersinking is only large enough to allow for metal 'stretch' - overly large recesses will do more harm than good, either by reducing the available surface area, and/ or allowing the transistor flange to be distorted.

Do **not** be tempted to tighten the screws as much as you can. They need to be tight enough so that there is good contact, but over tightening will distort the metal and will generally make the thermal resistance higher. The use of at least two washers is recommended, as this will help to distribute the pressure more evenly, and a spring washer is essential to stop the screws from loosening. If you can obtain them, 'cup' or compression washers are highly recommended, as they will maintain a consistent pressure, despite any movement ('flow') of the heatsink material. This is especially important with extruded aluminium heatsinks (i.e. most of those available) for long term thermal stability. Cast heatsinks are a completely different alloy, and distortion does not seem to be an issue.

If you use a mounting bracket and can feel any difference in temperature between the main heatsink and the bracket, there is thermal resistance present. More work might be needed to ensure that it is reduced to the minimum.

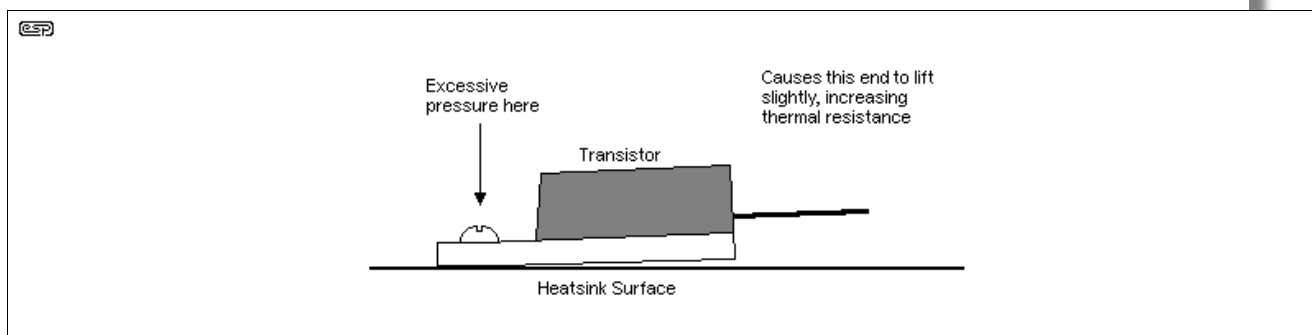
Remember too, that aluminium is a soft metal, and will tend to 'flow' when a consistent pressure is exerted. Over time, this may lead to the thermal resistance increasing, possibly to the point where the transistors overheat and are destroyed. For this reason, excessive tightening is never recommended, and the better the contact between bracket and heatsink, the less pressure is needed to ensure thermal resistance is kept to the minimum. Aluminium pop rivets can also be used, since they have the same rate of expansion as the heatsink and bracket (which are also aluminium), and will exert a consistent pressure which is not excessive. Care must be taken to ensure that the rivets are long enough to penetrate the heatsink, or the holding power is reduced to the point where it's slightly less than useless.



Note that if you are using rivets to attach a bracket, the bracket *must* be attached to the heatsink *before* the transistors are mounted. The shock from the rivetter may cause damage to the transistor die. *Never* use rivets to attach transistors to a heatsink or bracket.

### 13 - Flat-Pack Transistors

The TO-220, TOP-3 and other flat plastic packages have some real problems with thermal resistance, and their mounting arrangement does nothing to alleviate this - in fact the reverse is true. The problem is that the transistor junction is physically located more or less in the middle of the plastic encapsulated section, and the mounting is by a single tab, separated from the die by perhaps 10mm or so. This does not sound like very much, but the thickness of the tab is such that a significant heat difference can be measured between the two points.



**Figure 4b - One Result of Excessive Pressure**

When the device is mounted, the pressure of the mounting screw is concentrated in one spot, and may even cause the remainder of the transistor case to lift up from the heatsink slightly. When maximum heat transfer is required (which is most of the time), a very worthwhile improvement can be obtained by clamping the transistor to the heatsink with a section of flat aluminium bar (or even better, a channel or u-section). This should be thick enough to not bend when the screws are tightened, and also alleviates the requirement for insulating bushes and the like, since the normal mounting screws are not needed (the insulating washer is naturally still needed). Because of the pressure exerted directly above the transistor junction, thermal resistance is reduced by a very worthwhile margin, but care must be taken to



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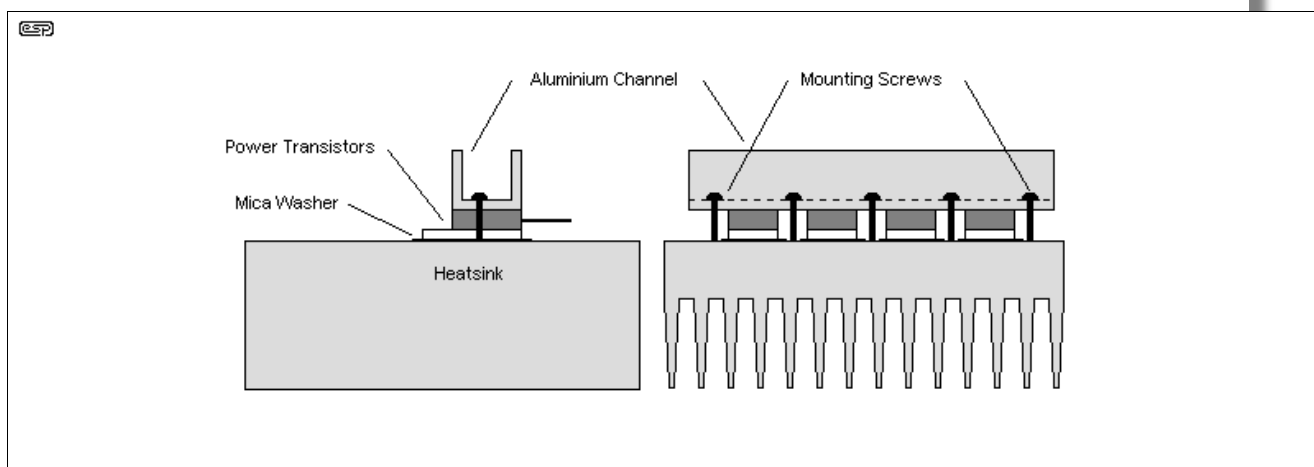
ensure that the screws holding the pressure bar are tightened evenly to ensure that all devices are firmly clamped. This method has the additional advantage that any deformation of the heatsink surface due to the pressure of the screw thread will be away from the device itself, enhancing long-term reliability.

See [Table 2A](#) for suggested mounting torque for different transistor cases.

A smear of thermal compound on the top of each transistor allows the pressure bar to become a part of the heatsink - not to the degree that a smaller sink can be used, but any reduction in thermal resistance is worthwhile, however slight.

Excessive pressure is a definite no-no, as you might be strong enough to crack the transistor case if you are not careful. This is one for those with some skill with machines and the like, since we are attempting to provide the absolute maximum allowable pressure on the cases, without the risk of damage - mechanical engineering stuff, basically.

The other advantage of this technique is that it allows the use of larger screws than would normally be possible, since they no longer have to be able to fit through the little hole in the insulating bush. This means more pressure and less thermal resistance, but only if the screws are tightened to a uniform torque so all devices have the same pressure exerted upon their cases.



**Figure 5 - Suggested Method For Mounting Flat-Packs**

Figure 5 shows the general idea, with four transistors mounted on the heatsink. On the left is a side view, with an 'end-on' view of the transistors on the right. This method requires that there is some free area on the PCB, since the transistor leads will normally be mounted directly into the board. The aluminium channel section can then perform another very useful function - supporting the circuit board so vibration or flexing does not fracture the transistor leads.

In addition, some amps need a heatsink for the driver transistors, and - there it is! I am surprised that this method is not more widely used in production amps (but then again, I only thought of it a couple of years ago, so maybe they just haven't thought of it yet). If this is the case, just remember where you saw it first. 😊

If you are a little wary, thinking that the mica washers might slip during assembly, use nylon screws to hold the transistors down loosely during assembly (don't tighten them too much, or they just snap off). These will hold everything in place while you go about the business of carefully tightening down the channel section screws. Remember that they should be all fairly even, and it is best to start from the centre, then work out towards each end to prevent any possibility of the channel section developing a buckle (however slight). If available, a torque wrench is very useful for tightening the screws (no, not the one you use for the car's cylinder head). 😊

#### 14 - Transistor Placement

Since heat rises by convection, one might think that mounting the transistors on the bottom of the heatsink might improve thermal performance. This is incorrect, and will actually cause the devices to run hotter. Aluminium is a good conductor of heat, and the idea is to ensure that the transistors are mounted in such a way that each has sufficient clearance from the others, and they are close to the



geometric centre of the sink.

This method ensures that the heat from each transistor has a similar mass of aluminium to disperse its heat into, ensuring that all devices have a similar and equal chance of dispersing the heat generated.

Convection (as we normally think of it) does not apply in a solid, since it requires molecular movement on a grand scale. In a solid, the molecules merely jiggle about more, but stay in the same place. Convection *only* occurs in a fluid, such as air or water.

For applications where a lot of heat energy has to be moved, a copper baseplate is useful. Copper has much higher thermal conductivity than aluminium, so a (thick) sheet of copper can be used as a 'heat spreader' to ensure that the heatsink is utilised to its utmost. The mating surface between the copper spreader and aluminium heatsink is every bit as critical as the transistor mounting. The surfaces must make perfect contact, and thermal grease or epoxy is used to fill the molecular air-gaps that would otherwise ruin the heat transfer. Needless to say, the copper has to be thick enough to carry the heat from the device(s) and spread it evenly across the heatsink itself.

Thermal mass (bulk metal) can be your friend for materials with good thermal conductivity, but it can also lull you into a false sense of security. If the dissipated power is variable or you don't test for long enough, you may find that the only thing that keeps the temperature within limits is the thermal mass. Apply power long-term, and you discover that the heatsink gets much hotter than expected. You need thermal mass, but you also need surface area to get the heat out of the heatsink and into its surroundings.

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## 15 - Heatsink Surface Area

There is only one thing on a heatsink that actually gets rid of heat to the surrounding air - surface area. The greater the surface area, the more heat will be disposed of. However, this is only the start, since fluid dynamics (in this case the fluid is air) also plays an important part. Make sure that you read the sections below on fin density and altitude effect - these are very important considerations in the overall design. Likewise, you can't reasonably expect to be able to shift a couple of hundred Watts longitudinally through 1mm aluminium plate - it won't happen. Well, it *will*, but the thermal gradient will be significant. The heatsink must be thick enough to carry the heat from the source to the fins.

Heat is lost to the air by two mechanisms, and both should be maximised for best performance:

- Conduction / Convection
- Radiation

Conduction (and/ or convection) requires that there is a continuous stream of air flowing past the fins of the heatsink, which means that the fins should be vertical if at all possible. Horizontally oriented fins will lose a vast amount of thermal transfer, since the air cannot flow through to the body of the heatsink. A fan solves this problem immediately, since air being blown onto the face of the heatsink has considerable turbulence, ensuring that there is always plenty of cool air at the surface of the heatsink.

Simple convection is not as effective (even for the same rate of flow of air), because of the 'laminar' flow of air (where the air at the surface of the heatsink moves slower than that further away). This effect can be easily seen on a windy day. If you stay close to a wall or other large area (lying on the ground works too), it will be noticed that it is less windy than out in the open. Exactly the same thing happens with heatsinks (but on a somewhat reduced scale). Creating turbulence is an excellent way to defeat this process, but this requires fans, and fans are noisy.

Radiation requires that the surface has the maximum emissivity of heat, and this means that its colour is important. Shiny gold anodised heatsinks might look great (if you like that sort of thing), but are hopeless at radiating heat. It's no accident that the radiator in a car, or the condenser on the back of a refrigerator is matte black - not chrome plated and shiny. Matte black heatsinks are the best for radiation, and will have a significantly better thermal resistance than any other. Use of paint is to be avoided however, unless kept very thin and even. A thick layer of paint acts as an insulator, reducing the ability of all those unwanted therms to get out into the air.

A point made by a reader is that to establish the *radiant* part of a heatsink is to enclose it in an imaginary box, and the outer surfaces of the *box* define the radiation area. This means that if the sides back (if exposed to free air) and just the tips of the fins are black, then that's all that's needed.



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Radiation between fins simply passes heat from one to another until thermal equilibrium is reached, but nothing is lost to the surrounding medium (air). Only the outer surfaces contribute to radiation loss. The transfer of heat between fins may not seem helpful, but it does help to ensure that the whole heatsink is at (close to) the same temperature. Remember that convection losses increase with temperature, so having the entire heatsink at an even temperature helps to maintain the highest convective losses possible.

One of the reasons that aluminium is so popular as a heatsink material is that it can be anodised (it's also comparatively cheap and is effective). Black dye is then introduced into the porous layer of aluminium oxide. This is far thinner than any coat of paint, and it is very effective, at least for the outer surfaces. Copper is actually a far better conductor of heat, but cannot be anodised, and its colour is such that it is a naturally terrible radiator. Such is life. Oxidised copper (kinda, sorta like anodising) is passably effective, but rarely used due to its cost. Table 3 shows the 'emissivity' of various surfaces. This is a measure of their ability to emit infra-red radiation (heat), and a figure of 1 is as good as it gets for a passive heatsink (i.e. no heat pumps or the like). This list is not exhaustive, but is a fair indicator for the most common surface treatments.

| Surface                  | Emissivity  |
|--------------------------|-------------|
| Polished aluminium       | 0.05        |
| Polished copper          | 0.07        |
| Rolled sheet steel       | 0.66        |
| Oxidised copper          | 0.70        |
| Black anodised aluminium | 0.70 - 0.90 |
| Black air-drying enamel  | 0.85 - 0.91 |
| Dark varnish             | 0.89 - 0.93 |
| Black oil paint          | 0.92 - 0.96 |

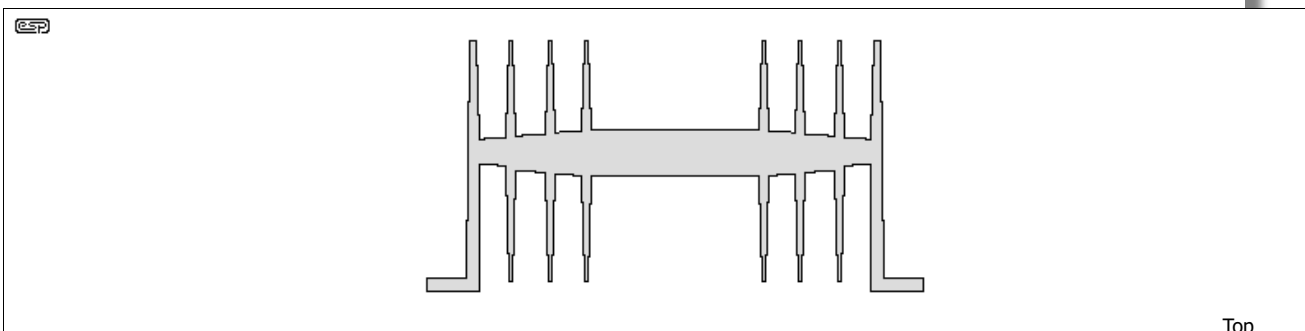
**Table 4 - Emissivity of Various Surface Treatments**



Emissivity refers to radiation, which is only a relatively minor (typically 25% or less) but still important means of dissipating the heat. Most heat is conducted to the air at the surface boundary, and although black oil paint has excellent emissivity, it will also insulate the fins from the air. Overall, as discussed above, about the best treatment is black anodising, but matte black automotive type lacquer is also very good (IMO) - provided it is applied as *thinly as possible*. Only the outer surfaces are involved in radiation - almost all internal radiation simply transfers heat from one fin to the other until thermal equilibrium is achieved.

Many high performance heatsinks have the surface of the fins ribbed, which further increases the surface area, so even a relatively small heatsink will have a radiating surface far greater than its moderate size would indicate, after all the individual fins, and the ribbing on the fins, is taken into account. Heavy application of black paint (for example) could fill the ribs, and reduce performance dramatically!

Ribbed fins don't usually make a great deal of difference with natural convection, because the airflow through the fins tends to be laminar and the air inside the ribs is effectively trapped, so doesn't move fast enough to be useful. Fan forced cooling can make full use of ribbing, provided that the airflow is turbulent. Unfortunately, turbulence also means noise, and that's often unacceptable for hi-fi applications where low noise (of all types) is desirable.



**Figure 6 - A Typical Heatsink Section**

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The simple heatsink in Figure 6, (100 x 100 x 50mm high), has a base surface area of 200cm<sup>2</sup>, and when the fins are added, this increases to 1000cm<sup>2</sup>. Note the way the base tapers away from the centre - this saves metal, but also provides maximum thermal inertia where the transistors mount, and matches the heat carrying capacity at the extremities to the expected heat flow. In reality, such a heatsink would have more fins, and if they were ribbed, the resultant final unit could have a total surface area of perhaps 1500cm<sup>2</sup>. This is the equivalent of a single plate over 270mm square, but will have far better performance because the thermal resistances are kept to a minimum by reduced distances. (Remember that both sides of the metal are in contact with the air, so the total surface area is double that expected.)

Note that when dealing with heatsink temperature, it is almost always referred to as temperature rise - the number of degrees above ambient that the heatsink will reach. If ambient temperature is lower than the 'standard' 25°C, so too will be the heatsink temperature (and vice versa, of course). Temperature rise is a constant in the equations - this is where we must be careful when the ambient temperature is high. A car in the sun can have an internal temperature of 50°C quite easily, so a temperature rise of 25°C will see the heatsink at 75°C.

The thermal resistance of a heatsink is determined by ...

- Its size
- The number of fins
- The total surface area - a ribbed fin design will be smaller than one using flat fins for the same thermal resistance
- Thermal resistance within the aluminium itself - the actual alloy used, its thickness, etc.
- The colour - matte black is best, and other colours are all worse
- The orientation of the fins - vertical is best, any horizontal position is worse
- Availability of unrestricted air flow
- Its temperature!

According to the 'quick and dirty' formula on Harry's Homebrew Homepage (the heatsink section seems to have gone), the thermal resistance is approximately equal to ...

$$\text{Thermal Resistance (R}_{\text{th}}) = 50 / \sqrt{A} \quad \text{Where A is the total surface area in cm}^2$$

Using this formula on the above heatsink (without ribbed fins), gives a thermal resistance of about 1.58°C/W or 1.29°C/W with the ribbed fins.

The formula can be turned around, so if you need (say) an R<sub>th</sub> of 5°C/W, you can get a rough idea with the following ...

$$\begin{aligned} \text{Area} &= (50 / \text{R}_{\text{th}})^2 & \text{So ...} \\ \text{Area} &= (50 / 5)^2 = 100\text{cm}^2 = 100 \times 100\text{mm} \end{aligned}$$

The same heatsink used in my heatsink calculator (see [Downloads](#) page) gave a thermal resistance of 1.12°C/W, assuming flat fins. I used a fin height of 50mm, depth of 100mm, and told it the heatsink has 8 fins. This assumed an ambient temperature of 25°C, and a maximum heatsink temperature of 40°C. This seems more realistic compared to manufacturer's data for similar heatsinks.

**Note:** Remember that 'ambient temperature' is that measured in the vicinity of the heatsink, and is not the temperature in the room. There are many reasons for an elevated 'ambient', and it's only the room temperature if there are no impediments to airflow, and the heatsink has *direct access* to room-temperature air. This may or may not apply, and design must be for the 'worst case' likely to be encountered. No sensible designer allows for user idiocy (like putting a power amplifier in a sealed cupboard), but you should allow for the 'ambient' temperature to be at least 5°C above the oft-quoted 25°C.

Manufacturers' ratings are not usually very specific, but we can safely assume that a given thermal resistance is the best obtainable, and probably assumes that the heatsink is in free air (no nearby casings, optimal fin orientation, etc) and in a 'standard' ambient temperature (typically 25°C). In most cases, a heatsink has to be used in a real-life situation, which means that you might need to add anything from 10% to 50% to the quoted figure, so a 1°C/W unit could end up as anything from 1.1°C to 2°C/W, depending upon its mounting, surrounding air temperature and any impediments to airflow.



Where possible, make sure that the heatsink is firmly attached to the case, which can make a reasonably thick aluminium chassis (for example) a significant part of the heatsink. This can improve performance markedly if good thermal bonding is used between case and sink. I do not recommend that you include the case in any calculations unless it is large compared to the heatsink, but if you do, do *not* count the inside of the case as part of the surface area. This is sealed (or semi-sealed), and the air will not have the free movement needed to remove the heat from the inner surfaces.

The last point in the above list is likely to cause some confusion - after all, thermal resistance is a constant based on the other factors, right? Wrong.

Let's imagine a heatsink with a heat load of 10W. After careful measurement, we determine that the temperature rise is 10°C (i.e. 10°C above ambient), so the heatsink has a thermal resistance of 1°C/W. We will almost certainly decide that a 10°C rise is probably a bit restrained, and in fact we can allow a temperature rise of 20°C, and we may decide to allow for a maximum 25°C ambient (not much use in outback Australia during summer, but it will do for this exercise).

Allowing an additional 10°C brings the thermal resistance down to 0.89°C/W, and if we decided that even a 35°C rise were acceptable (80°C heatsink temperature referred to a 25°C ambient), the thermal resistance drops further, to 0.73°C/W.

This apparently anomalous behaviour is actually all completely normal, and the heatsink is obeying the Second Law of Thermodynamics perfectly (feel free to look that up if you don't know it already).

In fact, the term 'thermal runaway' - where a transistor gets hotter so conducts more current, so gets hotter, so ... - is a misnomer in terms of physics. At some point, the system will stabilise, as it must. Unfortunately, this will probably involve temperatures well in excess of anything the transistor junction can tolerate, so it (or they) will be destroyed.

The heatsink discussed above has a thermal resistance of only about 0.45°C/W if a temperature rise of 200°C were acceptable (which it absolutely is *not*).

This is no different from any other situation where there is a difference in potential, pressure, or anything else. The greater the difference, the greater the flow rate. Now you know that the thermal resistance is *not* a constant - it is inversely proportional to the temperature difference between heatsink and ambient (this is a simplification, but it describes the behaviour fairly well). If heatsink and ambient temperatures are the same with a heat source applied, the heatsink has a thermal resistance of zero!

Thermal inertia can make it appear that's what we have, and a very brief test (a 1ms pulse test for example) will not cause much temperature rise, even with a small heatsink. Long-term use is what counts in most systems, so any test you do also has to last long enough for the heatsink temperature to stabilise. It won't be at ambient temperature though, unless you dissipate zero power!

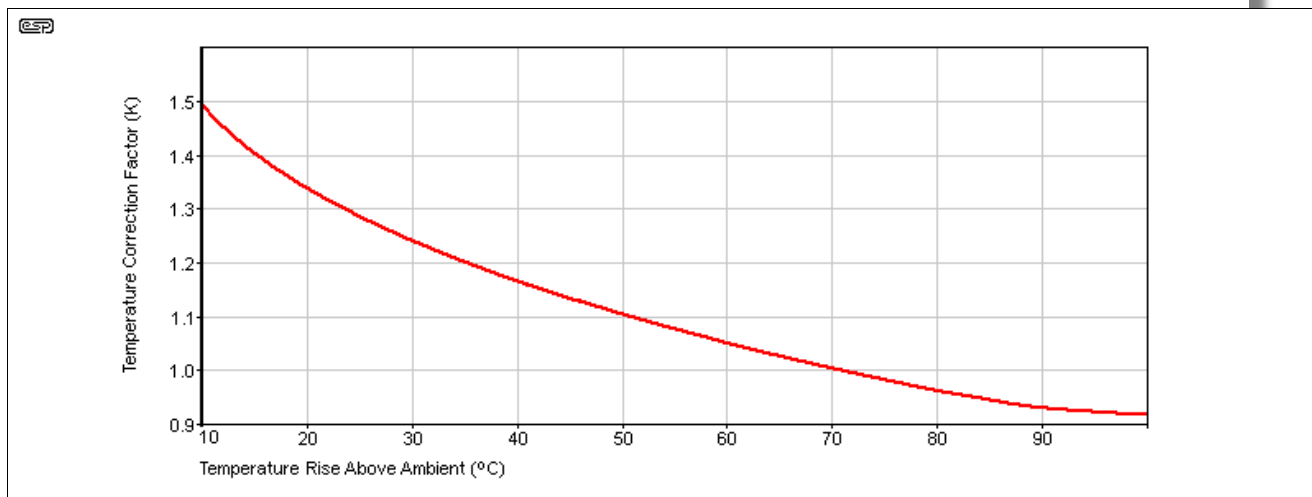


Figure 7 - Temperature Correction Based On Ambient Temperature

The above graph is *representative*, and assumes that the nominal thermal rating is based on a 70°C temperature rise. The 'K' value indicates the multiplication factor to get a heatsink of the desired rating. For example, with a 70°C rise, the heatsink gives its rated dissipation. If you wanted to reduce the

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temperature rise to (say) 30°C, the heatsink's thermal resistance will be higher by the 'K' factor. For example, a 1°C/W heatsink (at 70°C) is to be used at no higher than 30°C above ambient ...

$$\begin{aligned}\text{Heatsink (70°C)} &= 1^\circ\text{C/W} \\ 30^\circ\text{C rating} &= 1 \times 'K' (1.25) = 1.25^\circ\text{C/W}\end{aligned}$$

This is not exact, and may differ depending on the design of the heatsink. However, as a general *trend* it is useful. Remember that the rating is for temperature *rise* above ambient, so a 30°C rise means the heatsink is at 55°C with a 25°C ambient. Also, don't forget that 'ambient temperature' means the temperature near the heat generating surface, and *not* the air temperature in the room. An amplifier in a sealed cupboard will raise the internal temperature of said cupboard, possibly to a dangerous level! Any electronic product that generates heat requires proper ventilation, or it will probably fail if operated at its limits.

### 15.1 - Thermal Conductivity Of Various Materials

The thermal conductivity of the heatsink material determines how quickly heat can be dispersed from the transistor contact area(s) to the body of the heatsink. Thermal conductivity is simply the inverse of thermal resistance (often written as  $\theta$ ), so high conductivity means low resistance. Table 5 shows the thermal resistance of some materials. It is unlikely that too many people will opt for a diamond heatsink, but it's an ideal heat spreader for semiconductor dies that concentrate a large amount of heat in a very small area. However, for obvious reasons, it's not common. The figures below are mostly approximate, and apply for a TO-264/ TO-P3 package, having an area of 232 square mm (232 $\mu\text{m}^2$ ).

| Material/ Alloy                                  | Thermal Conductivity (W/m·K)                               |
|--------------------------------------------------|------------------------------------------------------------|
| Copper                                           | 385                                                        |
| Pure Aluminium                                   | 225                                                        |
| Aluminium/ 1100                                  | 218                                                        |
| Aluminium/ 6063                                  | 203                                                        |
| Aluminium/ 6061                                  | 167                                                        |
| Aluminium/ Cast                                  | 121                                                        |
| Brass                                            | 120                                                        |
| Iron                                             | 76                                                         |
| Aluminium Nitride                                | 320 ( $R_{th} = 0.01\text{K/W } 1\text{mm}$ )              |
| Graphite (avg) *                                 | 150 ( $R_{th} = 0.01\text{K/W } .5\text{mm}$ )             |
| Solder (60/40 Sn/Pb) *                           | 50 ( $R_{th} = 0.033\text{K/W } 0.2\text{mm}$ ) TO-262 SMD |
| Aluminium Oxide                                  | 30 ( $R_{th} = 0.1\text{K/W } 1\text{mm}$ )                |
| Silicone Pad #1 (10V/mm breakdown!)              | 25 ( $R_{th} = 0.062\text{K/W } 0.5\text{mm}$ )            |
| Silicone Pad #2 (150mm <sup>2</sup> sheet, soft) | 3 ( $R_{th} = 0.516\text{K/W } 0.5\text{mm}$ )             |
| Silicone Pad #3 (ceramic filled)                 | 1.2 ( $R_{th} = 1.3\text{K/W } 0.5\text{mm}$ )             |
| Silicone Pad #4 ('standard' thin)                | 0.9 ( $R_{th} = 0.612\text{K/W } 0.178\text{mm}$ )         |
| Kapton                                           | 0.2 ( $R_{th} = 0.387\text{K/W } 25\mu\text{m}$ )          |
| Nylon (Oven Bag)                                 | 0.2 ( $R_{th} = 0.155\text{K/W } 10\mu\text{m}$ )          |

**Table 5 - Thermal Conductivity Of Various Materials (\* Conductive)**

Update Dec 2023 - I have added a selection of TIMs (thermal interface materials) to the list, and calculated their thermal resistance based on a TO-264/ TO-P3P package, which have a metal surface area of 323 $\mu\text{m}^2$  (17 × 19mm). The TO-247 is smaller (231 $\mu\text{m}^2$ ) and the TO-220 is smaller again (~135 $\mu\text{m}^2$ ). The thermal *conductivity* is a constant, but thermal *resistance* is based on the area, thickness and thermal conductivity. Note that the area is that of the metal face of the transistor/ MOSFET, *not* the area of the thermal interface material! Solder is used with SMD devices, which are smaller than their through-hole equivalents. Tin/lead solder is actually a little worse than lead-free (>99% tin), but there's not much between them. Of course, solder is electrically conductive, so it doesn't provide electrical isolation.

While the figure for solder looks really good, it's conductive, and normally connects a transistor to a PCB with a very thin copper layer. The heatsinking that can be achieved is suitable for low power only, and even a fan won't help. Graphite is excellent (as are most other forms of carbon, including diamond), but as *graphite* it's conductive, and there is no insulation. Diamond has very high thermal conductivity and very high electrical isolation, but is not available as a transistor thermal interface



material (a pity, but for fairly obvious reasons).

To convert from thermal conductivity (W/(m·K) to thermal resistance (K/W or °C/W) use the formula ...

$$R_{th} = t / k \times A \quad \text{where ...}$$

t is thickness in metres  
k is thermal conductivity in W/(m·K)  
a is area in square metres

For any thermal interface material (TIM), your mounting technique (including the thermal grease you use) can only make the above figures worse, *never* better. Sorry.

For more info on the thermal conductivity of thermal interface material (TIM), see [section 11.3 - Thermal Conductivity](#).

For the heatsink, pure aluminium is very good, but is extremely difficult to machine or extrude, so it's thermal resistance is more of a reference value than anything else. Various alloys can have very different thermal resistance as shown in the table. Unfortunately, most suppliers neglect to tell us the alloy used. They may (or may not) disclose the temperature used to determine thermal resistance from the heatsink to ambient. This makes a significant difference, and high temperatures indicate a lower thermal resistance. Remember that if the heatsink is at the same temperature as the ambient air, the thermal resistance is infinite! (See 'main' Section 15.)

While cast heatsinks can look very nice, thermal conductivity is nowhere near as good as extruded alloys, so the base has to be much thicker to ensure that heat is transferred to the fins in a timely manner. High thermal resistance means that it takes longer for the heat to disperse, and the thermal gradient across a given distance is higher than for a material with lower thermal resistance.

Iron and/or mild steel is a poor conductor of heat, so if you thought that a sheet steel cabinet might help disperse heat you are in for an unpleasant surprise. Anyone who has worked with steel knows that the heat transfer is very slow ... one end of even a fairly short steel bar can be red hot, but you can still hold the other end in bare hands. It takes some time before the heat travels from the hot end of the bar to the 'cool' end. I have seen one attempt (which was a dismal failure) where a manufacturer of guitar amps thought that the steel chassis would aid heatsinking - it didn't, and their amps blew up with monotonous regularity.

Most power transistors use a copper heat spreader, because that helps to disperse the heat from the small die to the relatively large case far more effectively than simply attaching the die to the case itself. This is especially true with TO3 devices, since the case is made from steel. Aluminium TO-3 cases were used at one stage, but they quickly became unpopular (apparently) because of reduced reliability. No-one makes aluminium cased transistors any more, so that should tell you something. Many plastic power transistors use a copper backing and/or tab to ensure efficient transfer from the die to the case itself.

## 16 - Altitude

Since the density of air varies with altitude, so does the efficiency of a heatsink. I wonder how many consumer equipment makers take this into consideration? As can be seen from the table below, the effects are not insignificant.

| Altitude (Metres) | Altitude (Feet) | Derating Factor |
|-------------------|-----------------|-----------------|
| 0 (sea level)     | 0               | 1.00            |
| 1,000             | 3,000           | 0.95            |
| 1,500             | 5,000           | 0.90            |
| 2,000             | 7,000           | 0.86            |
| 3,000             | 10,000          | 0.80            |
| 3,500             | 12,000          | 0.75            |

**Table 6 - Altitude Derating Factors**

The altitude effect should be considered in all cases, as is evident. While the air temperature of an indoor environment is normally controlled and is not affected by the altitude change, the indoor air pressure does change with the altitude. Since many electronic systems are installed at an elevated altitude, it is necessary to derate the heat sink performance mainly due to the lower air density caused



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by the lower air pressure at higher altitude. The table shows the performance derating factors for typical heat sinks at high altitudes. For example, in order to determine the actual thermal performance of a heat sink at altitudes other than sea level, the thermal resistance values read off from the performance graphs should be divided by the derating factor before the values are compared with the required thermal resistance.

Example: A 1°C/W heatsink would become 1.16°C/W at an altitude of 2,000 metres, or 1.25°C/W at 3,000 metres.

## 17 - Fin Density

Although it would seem reasonable to assume that the more fins per unit area the better, this is not always the case. Closely spaced fins will not be able to dispose of the heat well in a standard convection (i.e. not fan forced) heatsink. This is partly because of the airflow that will actually be able to establish itself within a confined space, and partly because the fins will tend to radiate much of the heat to adjoining fins. This helps to stabilise the temperature, but does little to dispose of the heat to the atmosphere.

The maximum distance between fins is dependent on the depth or height of the fins - deep finned heatsinks will need more space between adjacent fins than a shallow design unless fan cooling is used. As shown in the following table, the minimum spacing is determined by fin depth and airflow. When heatsinks are fan-cooled, the fin density can be increased (sometimes dramatically). Very close spacing, cross-fins (often created by separate extrusions force-fit into the main heat spreader) and other techniques are used to get very high fin density. As the fin density increases, so does the needed power for the fan. A fan is of no use if it can't force enough air between the fins. High-density heatsinks can provide thermal resistances below 0.1°C/W, but that will typically require airflow of at least 200m<sup>3</sup>/minute. This class of heatsink is expensive - you'll be looking at over AU\$500 (as of late 2023) for a suitable candidate - some are much more.

| Fin Height (millimetres) | 75               | 150 | 225 | 300 |
|--------------------------|------------------|-----|-----|-----|
| Airflow (metres / sec)   | Fin Spacing (mm) |     |     |     |
| Natural convection       | 6.5              | 7.5 | 10  | 13  |
| 1.0                      | 4.0              | 5.0 | 6.0 | 7.0 |
| 2.5                      | 2.5              | 3.3 | 4.0 | 5.0 |
| 5.0                      | 2.0              | 2.5 | 3.0 | 3.5 |

Table 7 - Fin spacing (in mm) versus flow and fin length

The average performance of a typical heat sink is linearly proportional to the width of a heat sink in the direction perpendicular to the airflow, and approximately proportional to the square root of the fin length in the direction parallel to the flow. For example, an increase in the width of a heat sink by a factor of two would increase the heat dissipation capability by a factor of two, whereas doubling the depth or height will only increase the heat dissipation capability by a factor of 1.4. Therefore, if the choice is available, it is beneficial to increase the width of a heat sink rather than the length of the heat sink or the height of the fins. Also, the effect of radiation heat transfer is very important in natural convection, as it can be responsible of up to 25% of the total heat dissipation. Unless the heatsink is facing a hotter surface nearby, it is imperative to have the heat sink surfaces painted or anodised black to enhance radiation.

The reason for the difference is simply that as the height is increased, the air at the top of the heatsink is hotter than that entering at the bottom. If the fin depth is increased, there is more mutual radiation between fins, and as the spacing is reduced, mutual radiation increases again. Airflow is also restricted because of the smaller physical area for air to pass, since more of the available space is occupied by the heatsink itself. Fan cooling removes many of these restrictions, but the fan *must* be powerful enough to maintain an air flow rate sufficient to move hot air out as quickly as possible.

One class of heatsink uses what are called 'skived' fins<sup>1</sup>. These are usually both very thin (typically around 0.5mm thick) and close together. A sharp blade (with a great deal of force) is used to shave a thin layer of material from one side of the substrate (aluminium or copper), and it's either produced as a curled fin or bent vertically to form closely spaced fins. This type of heatsink is not suitable for convective cooling, and a fan is usually mandatory.

<sup>1</sup> Skive: avoid work or a duty by staying away or leaving early; shirk, or To shave off the top surface of material to make it less thick.



Predictably, the term 'skive' in the heatsink context is the second definition. You may find examples of skived fin heatsinks in your PC for CPU or GPU cooling. They are not an everyday sight, but I do have a couple that were (very inappropriately) used in Chinese made powered speaker box amplifiers. They are common in industrial systems where extremely good cooling is needed. An example of the process can be seen on [YouTube](#). There's a great deal of info on both the process and applications on the interweb of course, so if you want to know more, just do a search for 'skived heatsinks'.

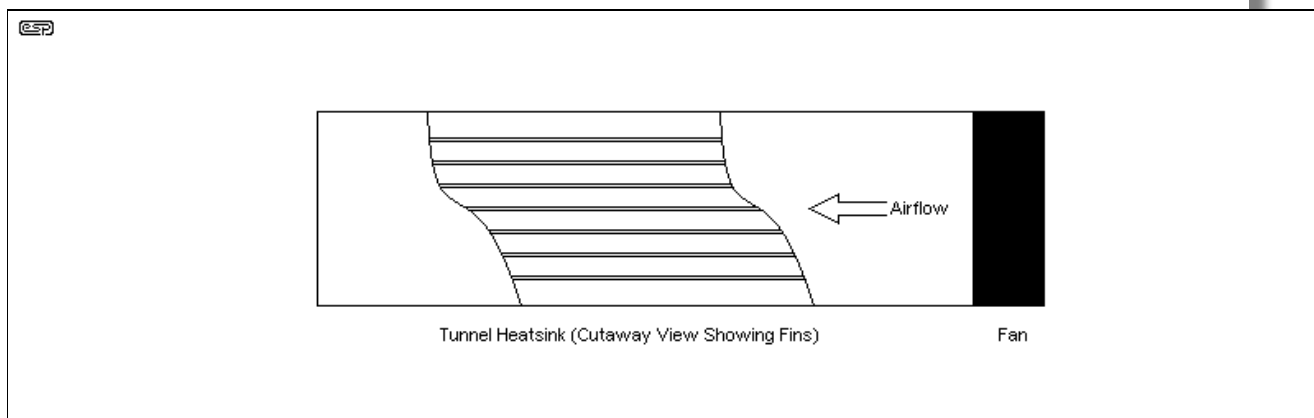
Another process used for some complex shapes is forging - the aluminium is shaped using special dies and immense pressure. As with skived heatsinks, there are videos available that show the manufacture of forged heatsinks. In some cases it may be difficult to work out just how a particular shape has been created, and if it's obviously impossible for it to have been extruded (most heatsinks are made using this linear process), then it's entirely possible that it was forged. Like skived heatsinks, forged products will almost always require fan cooling because the fin density is too high for natural convection.

## 18 - Fan Cooling

Forced-air cooling using a fan is very common. Most high-powered PA amplifiers utilise fan cooling, and it's also seen in some multi-channel amps used for home theatre. Completely internal fans are used by many manufacturers, having a filter on the outside of the amp where it's accessible for cleaning, although it won't get cleaned by many users anyway. It seems that owners and/or users object to mundane chores like cleaning filters, or perhaps they just forget.

There is no doubt that even a modest heatsink can dispose of a prodigious amount of heat with forced cooling. This allows amp manufacturers to keep size and weight to the minimum, but still provide proper cooling for the power transistors. Naturally, all the other points raised above are still vitally important. The task is to get the heat from the transistor junction and dissipate it into the atmosphere, with the greatest efficiency possible. The effectiveness of heat removal depends on airflow and turbulence - both should be maximised for optimum cooling.

In general, we'd should aim for a fan that provides at least 1m<sup>3</sup>/minute (35CFM - cubic feet/minute). More is better, but the fans themselves get pretty powerful and noisy. The fan must also be able to overcome the static pressure generated as it tries to force air through the heatsink. Close fin spacing will increase the pressure needed for a given airflow. As with most things, there are trade-offs that you need to deal with, and sometimes only a physical test under controlled conditions will tell you if the system will work or not.



**Figure 8 - Correct Airflow Through A Tunnel Heatsink**

At first glance, you might think that it doesn't matter much whether the fan sucks or blows air into the tunnel. In reality, there is usually a big difference, with blowing (as shown) giving much better cooling (but also much higher noise). The reason is simple ... the air leaving the fan blades is turbulent, and it swirls around vigorously as it leaves the fan blades. This allows the airflow to 'scavenge' otherwise non-moving air from against the fins. The fins therefore get a continuous supply of cool air which aids heat removal. As noted above (in Fin Density), some heatsinks are specifically designed for forced-air cooling, and they are completely unsuitable for convection cooling.

Remember that the effectiveness of a heatsink depends on the temperature difference between the heatsink itself and the adjacent air. If the air is warm (right against the fins) then the heatsink must run hotter than it would with cooler air against the fins. This is a simple relationship, and determines the

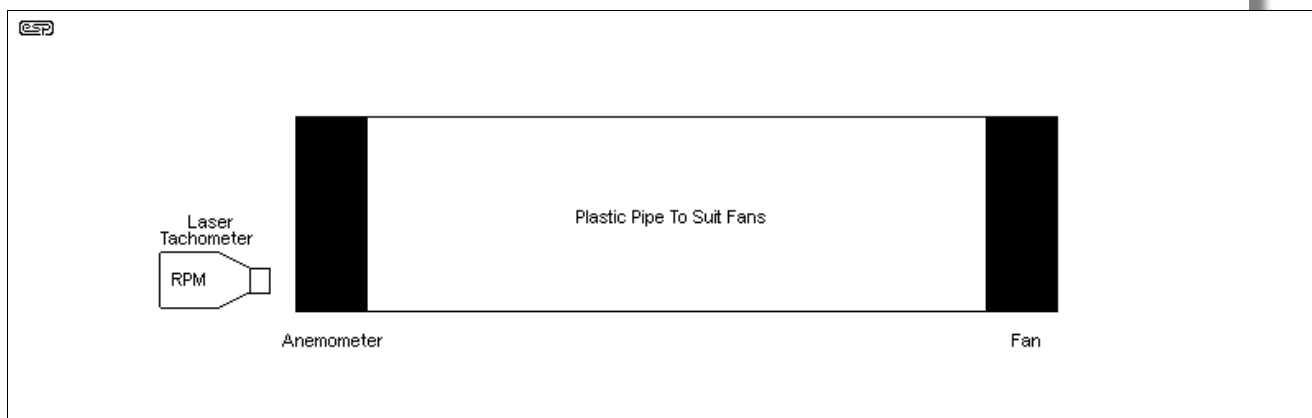


thermal rating of any heatsink.

Should the fan be connected so it *sucks* air into the tunnel, the airflow will be mostly laminar - moving fastest in the centre, with comparatively little movement at the surfaces of the fins where it's needed most. Without the turbulence that stirs up the airflow and making laminar flow impossible, the performance is reduced dramatically. The same applies to a conventional heatsink with a fan attached to the outside of the fins. The heatsink temperature difference between blowing and sucking can be 10-15°C or more, depending on the heat load <sup>[7]</sup>.

The golden rule of forced air cooling is that you want (and need) the greatest airflow and turbulence possible, so the fan should *always* blow air onto the heatsink. *Never* set up a fan to suck air across the heatsink, because as a method of cooling ... it sucks 😊.

As an experiment, I set up the arrangement shown in Figure 8. One fan was powered, and another was used as an anemometer, with a reflective strip on one blade. I used a laser tachometer to measure RPM when the powered fan was set to blow or suck air through the pipe. You would probably expect that there'd be no difference, but in fact it was even greater than I expected - I fully anticipated the result, but it was more pronounced than I thought it would be.



**Figure 9 - Test Setup For Airflow In Tube**

When the fan was blowing air into the pipe, the second 'anemometer' fan was driven to 1,230 RPM, and the anemometer would still turn even when 25mm from the end of the pipe. When the powered fan was reversed so it *sucked* air into the pipe, the anemometer only managed 860 RPM, and it would stop when only a few millimetres from the end of the pipe. This is a big difference, and shows that a blowing fan not only creates higher turbulence, but also pushes more air. The fan used as an anemometer was reversed along with the fan, because it spins more happily in one direction than the other.

I also verified that airflow with a sucking fan is highly concentrated in the centre of the pipe. A thin piece of tissue paper was used, and it migrated to the centre. When forced against the side of the pipe, it was easy to get it to stop fluttering completely. This shows that the air against the surface is almost still - proof positive of laminar airflow. On the downside, when the fan blows *into* the pipe, it is much noisier, primarily due to turbulence. In general, it's safe to say that a bit of extra noise is better than having an amplifier fail due to overheating.

If you were to look at the specifications for fan-forced heatsinks, you'll find that the thermal resistance reaches a plateau with heatsinks longer than ~200mm. For best cooling, use two 100mm long heatsinks (each with its own fan) rather than a single 200mm model. While this arrangement will cost more, its performance is far superior.

If a fan-forced heatsink is too long, the temperature rise at the far end (away from the fan) can be considerably higher than at the near end. Turbulence diminishes with distance, and without turbulence you have an under-performing heatsink. If you use a thermal cutout, it has to be mounted well clear of the fan end, or the temperature measured will be very optimistic. Two short fan assisted heatsinks (with a fan on each) will outperform a single heatsink of the same total length. The further away from the fan you get, less turbulence is available and the airflow starts to become laminar again. This reduces the effectiveness surprisingly quickly.



The conclusion is unmistakable - fans should *blow* air into a tunnel heatsink or onto an exposed heatsink. Sucking not only doesn't create the turbulence we need for effective heat removal, it doesn't

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even move as much air where it's really needed! Does anyone stand *behind* a pedestal fan to keep cool on a hot day? Case closed.

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## 19 - Water Cooling

I was going to stay away from this completely, but it is worth at least a small section. Water just happens to be the best heat removal medium known, with a specific heat of 4.1813 (J/g·K), it requires more energy to raise a gram of water by 1°C (or 1 Kelvin) than any other material (other than hydrogen, ammonia or liquid lithium, but they're not even remotely useful for our purposes).

If extremely high power is the goal (and a bit of plumbing is Ok), a water cooled heatsink is ideal. Provided the thermal resistance from junction to heatsink is minimised, a minuscule heatsink with only a moderate water flow will remove prodigious amounts of heat. Although uncommon for audio amplifiers, water cooling has been used for many years for cooling high power radio transmitters and the like.

However, it must be admitted that few audiophiles will go to the trouble of installing special plumbing to cool their amplifiers - it would be cheaper to put them in another room and use fan cooling, and a lot more convenient.

Having said that, the use of water cooling is gaining in popularity, and quite a few commercial H2O cooling systems are available for computer processors. As component density increases in the ICs, it becomes harder to get the heat out efficiently, so expect to see that side of the market expand in the next few years. While it must be admitted that water cooled heatsinks are the best possible choice for amplifiers, this is not an area that can be expected to grow, since it is just too expensive and unwieldy to implement. Finding a silent running pump is another hurdle, and although the plumbing is not at all difficult (it's all low pressure), it still has considerable nuisance value - the cables in a typical system are bad enough, let alone having a plumbing system that must be 100% watertight. Of course, then there is the radiator, pump and fan to be considered - they have to live somewhere, and can be expected to have a SAF (Spouse Acceptance Factor) of perhaps -20dB ("I don't know what you're planning to do with *that* horrible looking thing, but if it comes in here, I'm moving out!" - a fairly typical response, I would suggest).

Still, I may do a water cooled heatsink project at some time, just for the fun of it 😊.

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## 20 - Heat Pipes

Heat pipes used to be uncommon, but are now popular with high-end PCs. They will not be discussed at any length in this article because the 'cool' surface is quite small - just big enough for a processor IC. They vary in price, with some being surprisingly inexpensive, and are most commonly used where a hot component must be cooled in a confined space. The heat pipe is actually two heatsinks, joined by a pipe containing a refrigerant. Natural convection is used to move the refrigerant from the hot area (the component to be cooled) and the main heatsink, which may be mounted some distance away.

For those who are interested in this technology, there are quite a few articles on the web that discuss heat pipes, and I suggest a web search to locate them. These have become quite popular for cooling computer CPUs and GPUs (graphics processing units) and are often fairly economical. However, the cooling surface is too small to be useful for an amplifier or power supply. You could use (say) a pair of them, thermally connected to a copper bar with your transistors, but that's likely to be mechanically messy and difficult to achieve in the confines of an amplifier chassis.

There are some newer versions of the same basic principle now available. These make use of 'micro-channels' to dramatically increase the surface area at the hot end, and allow the use of more environmentally friendly coolants. However, it is to be expected that these will be expensive, and outside the budget of most home constructors.

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## 21 - The Bottom Line & Final Thoughts

As can be seen from the above, for a given heatsink, the most critical part of the whole thermal resistance equation is often the transistor itself. If the thermal resistance of the active devices can be reduced, the demands on the heatsink are lessened, resulting in a pleasing increase in reliability.

The thermal resistance from transistor case to heatsink is very important, and proper mounting



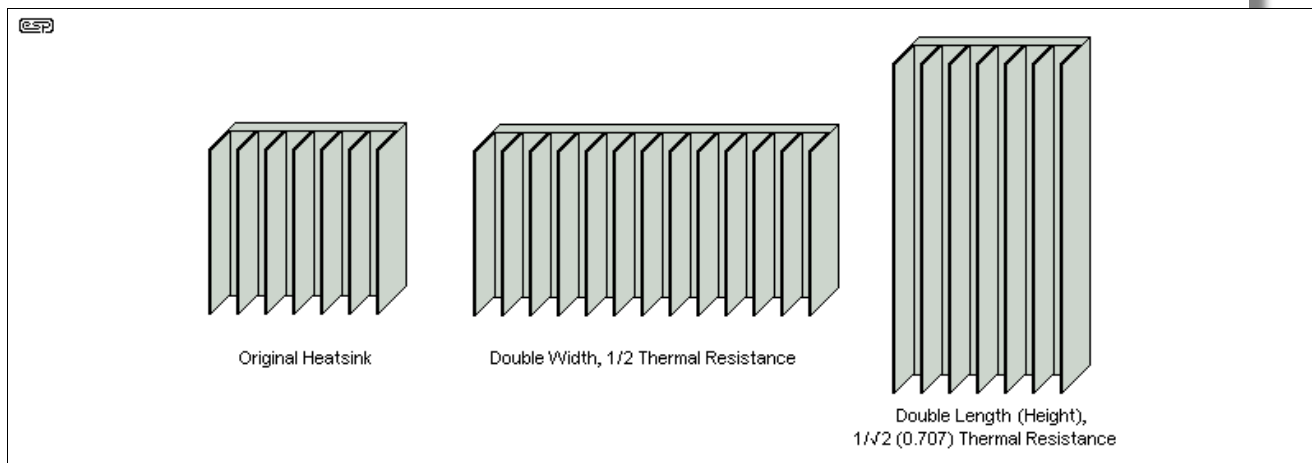
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technique can result in a significant improvement over the "just slop some grease on and bolt it down" approach. Selecting the right insulator can reduce thermal resistance dramatically, and even the choice of thermal compound (grease) can make a measurable difference.

If these two thermal resistances can be reduced enough, it may even be possible to make the heatsink itself somewhat smaller than would otherwise be the case for the same (or even lower) transistor operating temperatures. The whole process is a science - 'art' only comes into play to determine the aesthetics, which (to the product stylist) is considered the most important. It's not. There's no point having a product that looks wonderful but blows up when called upon to do any real work.

Always remember that manufacturers' data on heatsinks is under ideal conditions, and is usually measured at a temperature of 50 to 80°C above ambient - although unfortunately this is rarely stated. Hence the need for testing so that you know exactly what the heatsink can do under realistic conditions.

And a final thought? ...



**Figure 10 - Making Your heatsink Larger**

The above drawing shows what happens if you make a heatsink larger, and natural convection cooling is assumed. If the heatsink is made double the width, the thermal resistance is halved as you would expect - assuming of course that the heat sources are spread out to make use of the area. However, should you decide to make the heatsink twice as *long*, the thermal resistance is reduced by the inverse of the square root of the increase - in this case  $1/\sqrt{2}$  (0.707). To get half the thermal resistance, the heatsink would need to be 4 times longer.

Note that this apparently odd behaviour is quite reasonable, because the air that enters at the bottom of the fins passes the remainder of the heatsink gathering heat all the while. Since the air will be slightly hotter (at the top) than with a shorter heatsink, thermal transfer is reduced due to the higher (localised) ambient temperature.

This does *not* apply if the heatsink is fan cooled, provided the air velocity is high enough to expel the heated air before its temperature has risen too much. As always, the golden rule of fan forced cooling is more air, more air and more air (but not necessarily in that order). For maximum effect, the fan must *blow* air onto the heatsink fins.

As you have learned, heatsink design is not trivial, nor is the understanding of the physics behind thermal transfer, fluid dynamics, or any of the countless other things that affect performance. However, it's not that hard once you have the information you need and a few pointers that I hope have been helpful.

## 22 - Other Heatsink Information

I have included a heatsink calculator spreadsheet, which appears to work quite well. Go to the [Downloads](#) page to get a copy, or you can just [click here](#) for a download.

A very useful link for those who are put off by the cost of large heatsinks:



[Heatsinks](#) Very good drawings, showing the complete process for building a heatsink. The page is nov.

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hosted by ESP, as it had disappeared from its original host.

A most excellent document has found its way into my clutches! It is an application note from International Rectifier ([irf.com](http://irf.com)) and I now have a link to it [here](#) (on the ESP website) for your convenience. Although it refers to a specific case style, the information is generic enough to ensure that you have a better all-round understanding of the subject.

Another excellent document that describes mounting methods and techniques can be found at the [On Semiconductor](#) website. This describes the subject in considerable detail. I was made aware of this document after the article was written, and it was not used as a reference.

**Please Note:** With all articles of this nature, there will be some information that appears to be in conflict with other data you will see in other application notes or publications. This is perfectly normal, as no two engineers will ever be in complete agreement, but the basic data correlates very well, for the most part with only a few minor points of difference. It's inevitable that there will be differences, but in many cases it's just semantics - people write differently, even in engineering.

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### 23 - References:

1. General Electric Transistor Manual
  2. National Semiconductor Voltage Regulator Handbook
  3. Jaycar Electronics Engineering Catalogue
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  6. Aavid Thermal Technologies, Inc. Laconia, New Hampshire
  7. [Fan cooling: intake or exhaust? - 'halfgaar'](#)
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  9. Heat Sink Design Facts & Guidelines For Thermal Analysis - Wakefield-Vette Technical Brief
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**Main Index**



**Articles Index**

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Change Log: 16 Jun 1999 - Initial publication./ Aug 99 - new Figure 1 and explanation of electrical analogue./ Jan 00 - Corrected several errors, added links for additional info and more info on emissivity etc./ Feb 00 - 'additional thoughts' (Geoff Moss contribution)/ Apr 00 - info on heatsink calc spreadsheet./ Aug 00 - Added an-1000.pdf./ Feb 01 - info on mica 'splitting'./ Feb 02 - fin density and altitude effect, plus other small additions./ May 02 - corrected link to Harry's Homebrew page./ Sep 07 - page re-format./ Mar 2013 - section 18 (moved other sections down)/ May 2016 - phase change, re-drew surface imperfection drawing./ Jul 2019 - added info on simulation./ Nov 2019 - added skived heatsinks./ Jul 2020 - added Figure 7 and text./ Oct 2020 - added more info on radiation losses./ Feb 2021 - included info on  $W/(m \cdot K)$ . Mar 2021 - Added oven bag info (yes, really). Feb 2022 - minor update. Dec 2023 - included more info on thermal conductivity and the conversion formula./ May 24 - added further comment on heatsink size (intro)/ Jan 25 - added aluminium oxide & nitride (Section 15.1).



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